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A PROJECT REPORT ENTITLED
DYNAMIC CONTEXT-AWARE TRIES FOR KNOWLEDGE
GRAPH-CONSTRAINED LARGE LANGUAGE MODEL GENERATION

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DECLARATION

We, BERNARD KIRK ADJANOR KATAMANSO, ERICA AMONOR, JOSEPH OSEI NYARKO and JESSICA AFUA ETORNAM NSAFOAH, declare that this project work is our own original work. It is being submitted for the degree of Bachelor of Science in Computer Science and Engineering at the University of Mines and Technology (UMaT), Tarkwa. It has not been submitted, either in whole or in part, for any degree or examination at any other university.

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ABSTRACT

Knowledge Graph constrained decoding eliminates hallucination in large language model (LLM) reasoning by restricting output to verified paths. The leading approach, Graph Constrained Reasoning (GCR), achieves zero hallucination on Knowledge Graph Question Answering (KGQA) benchmarks by encoding all structurally valid paths in a static trie. The constraint is absolute but indifferent to the question: a three-hop query with twenty out-degree relations admits roughly 24,000 paths, and the LLM must navigate among them using the same internal knowledge that causes hallucination. We introduce Dynamic Context-Aware Tries (DCA-Trie), a dynamic constrained decoding framework that prunes the valid path set during generation using a TypeOracle, a lightweight symbolic module that checks two ontology gates per candidate triple. On WebQSP, DCA-Trie removes 85.5% of semantically irrelevant paths while maintaining a false negative rate below 5%. We further introduce the Semantic Irrelevance Ratio (SIR), a metric that separates constraint quality from answer accuracy, enabling independent evaluation of what the oracle filters versus what the LLM actually uses. SIR reveals a non-monotone relationship between constraint tightness and answer correctness: tighter constraints reduce the search space but do not improve Hits@1. We identify four mechanisms that drive this decoupling and show that the result holds across ablation conditions. DCA-Trie and SIR together provide the first framework for independently evaluating constraint quality and answer accuracy in knowledge graph-constrained decoding.

DEDICATION

Dedicated to our parents. For every sacrifice we didn't see and every kindness we never had to ask for. Thank you for everything.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
BFS	Breadth-First Search
CoT	Chain-of-Thought
CWQ	ComplexWebQuestions
DCA-Trie	Dynamic Context-Aware Trie
DoG	Decoding on Graphs
FNR	False Negative Rate
FSM	Finite State Machine
GCD	Grammar-Constrained Decoding
GCR	Graph-Constrained Reasoning
GNN	Graph Neural Network
GPU	Graphics Processing Unit
KG	Knowledge Graph
KGQA	Knowledge Graph Question Answering
LLM	Large Language Model
NLP	Natural Language Processing
QA	Question Answering
RAG	Retrieval-Augmented Generation
RLHF	Reinforcement Learning from Human Feedback
SBERT	Sentence Bidirectional Encoder Representations from Transformers
SIR	Semantic Irrelevance Ratio
SQL	Structured Query Language
VRAM	Video Random Access Memory

CHAPTER 1

INTRODUCTION

1.1 Background

Despite their fluency, large language models (LLMs) are susceptible to hallucination, generating coherent and plausible text that is factually wrong (Ji *et al.*, 2023; Huang *et al.*, 2025). Autoregressive language models generate text by sampling tokens from a learned distribution conditioned on prior context, with no mechanism for grounding outputs against external knowledge sources during decoding (Lewis *et al.*, 2020). Errors introduced at one decoding step can propagate to subsequent steps, and in knowledge-intensive tasks such errors manifest not as stylistic infelicities but as factual inaccuracies. This makes factuality a central limitation of LLMs in precisely the settings where correctness is most critical.

A knowledge graph (KG) stores facts as triplets (entity, relation, entity) and supports multi-hop reasoning over verifiable connections (Bollacker *et al.*, 2008). Benchmarks such as WebQSP (Yih *et al.*, 2016) and ComplexWebQuestions (Talmor and Berant, 2018) test whether a system can trace these connections to answer natural language questions. Unconstrained LLMs, relying on parametric memory, frequently produce plausible but hallucinated reasoning trajectories over such graphs (He *et al.*, 2021; Lan *et al.*, 2022).

Retrieval-Augmented Generation (RAG) mitigates hallucination by injecting retrieved evidence into the context window (Lewis *et al.*, 2020; Izacard and Grave, 2021). However, retrieved evidence functions as a soft prompt rather than a hard constraint on generation: models can disregard relevant retrieved context or generate claims unsupported by the retrieved material when internal parametric knowledge overrides external evidence (Asai *et al.*, 2024).

1.2 Problem Statement

Constrained decoding takes a harder line by applying a validity mask over the output vocabulary. At each step, a constraint oracle forces the LLM to generate only tokens that correspond to valid paths in the KG (Willard and Louf, 2023; Geng *et al.*, 2023). Graph-Constrained Reasoning (GCR) (Luo *et al.*, 2025) achieves this with a pre-computed KG-Trie used as the oracle during beam search. The guarantee is structural in that, every generated token maps to a real graph edge. The problem is that the trie is built before decoding begins and never changes. It cannot adapt to the semantic intent of the question or the evolving reasoning chain, hence the LLM must still use its own judgment to filter out structurally valid but irrelevant paths (Li *et al.*, 2025; Luo *et al.*, 2025).

Decoding on Graphs (DoG) (Li *et al.*, 2025) addresses this rigidity by expanding the trie dynamically during decoding. But DoG admits all topologically adjacent nodes without semantic filtering. The result is unnecessary computational overhead and continued vulnerability to off-target paths.

These gaps are the gaps that DCA-Trie is designed to close. By conditioning the valid token set on three inputs, the knowledge graph, the input question, and the partially generated reasoning chain, DCA-Trie combines structural constraints with dynamic semantic filtering, aiming to reduce constraint permissiveness without sacrificing the faithfulness guarantees that make constrained decoding valuable in the first place.

1.3 Aim and Objectives

The aim of this project is to design, implement, and evaluate a Dynamic Context-Aware Trie (DCA-Trie) framework for knowledge graph-constrained LLM generation. By conditioning each decoding step on the knowledge graph, the input question, and the partial reasoning chain to improve multi-hop KGQA accuracy while maintaining 100% structural faithfulness to the underlying knowledge graph.

The specific objectives of this project work are to:

- i. Characterize the constraint permissiveness problem in existing constrained-decoding approaches to KG-grounded reasoning;
- ii. Design a semantic relevance scoring mechanism and a constraint oracle for filtering structurally valid but semantically irrelevant paths;
- iii. Implement a static filtering variant that applies the oracle at trie construction time, and evaluate its impact on false negative rate;
- iv. Implement a dynamic variant in which the trie is updated during decoding, conditioned on the question and the partially generated reasoning chain; and
- v. Evaluate both variants against relevant baselines on standard knowledge-graph question answering benchmarks, and assess whether observed patterns generalize across benchmarks.

1.4 Tools and Facilities Used

1.4.1 Facilities

The following facilities were utilized in the execution of this project:

- i. NVIDIA GeForce RTX 4090 GPU (24 GB VRAM) for all model inference and constrained decoding experiments.
- ii. University library and online academic databases (ACM Digital Library, arXiv, Google Scholar, ACL Anthology) for literature review and citation verification.
- iii. Shared team repository hosted on GitHub for version control, experiment logging, and reproducibility of all reported results.

1.4.2 Software Tools

Table 1.1 presents the software tools used in this project.

Table 1.1 Implemented Software Tools

Tool	Use in Project	Rationale
Python / HuggingFace Transformers	Model inference pipeline for Llama-3.1-8B, including the constrained decoding callback	Standard tokenization and generation utilities for the released GCR checkpoint
GCR Framework	Primary baseline; KG-Trie construction and graph-constrained decoding	Provides the trie-based decoding mechanism DCA-Trie extends
PyTorch	Deep learning framework underpinning inference and decoding logic	Supports SDPA attention and bfloat16 inference
marisa_trie	KG-Trie construction and prefix-based token lookup	$O(1)$ prefix lookup for the valid-token mask at each decoding step
NetworkX	Graph construction and path enumeration from question entities	Builds directed graphs from Freebase triplets
Freebase / WebQSP / CWQ	Knowledge graph source and KGQA benchmarks	Standard benchmarks for comparison with prior work
NVIDIA RTX 4090 (24GB)	GPU for all inference experiments	Sufficient VRAM for bfloat16 inference of an 8B-parameter model
VS Code, Git / GitHub	Development environment and version control	Coding, version tracking, and reproducibility

1.5 Scope and Delimitation

This project targets the permissiveness problem in GCR and DoG: existing constraint oracles admit every structurally reachable path regardless of relevance. Evaluation is restricted to multi-hop KGQA over Freebase benchmarks (WebQSP and CWQ). All experiments use GCR’s publicly released Llama-3.1-8B checkpoint (Luo *et al.*, 2025); no fine-tuning is performed. Generalization to other KGs or non-KGQA tasks is identified as future work. All code and evaluation datasets will be made publicly available.

1.6 Organization of the Project Report

The rest of the report is organised as follows. Chapter 2 reviews literature on hallucination, knowledge graph question answering, and constrained decoding. Chapter 3 presents the research methodology, formalising DCA-Trie and its two variants. Chapter 4 reports results on WebQSP and CWQ, including ablation and error analysis. Chapter 5 presents conclusions and recommendations for future work.

CHAPTER 2

LITERATURE REVIEW

While large language models (LLMs) are good at many tasks, their use of autoregressive generation makes them likely to produce fluent but incorrect information (Ji *et al.*, 2023). For knowledge-intensive question answering, this unreliability is a major roadblock as the ability to verify reasoning against external facts is essential. To tackle this issue, recent studies have focused on grounding models in structured external facts through knowledge graphs. Still, keeping a model strictly aligned with these graphs during generation, without losing its natural flow, is a difficult challenge.

This literature review looks at the current research on these challenges and examines how knowledge graph constraints might help ensure factual accuracy during multi-hop reasoning. This chapter outlines the connections between language model hallucination, multi-step validation, and structured grounding. By assessing current strategies to reduce errors, from improved prompting to fixed decoding, this chapter shows where soft-grounding methods do not meet expectations, highlighting the specific technical gaps this project addresses.

2.1 Large Language Models and Hallucination

Modern LLMs are built on the Transformer architecture (Vaswani *et al.*, 2017), using scaled dot-product self-attention to weight token importance regardless of positional distance. The field has largely moved toward decoder-only models with causal masking, where each token attends only to preceding tokens. The Llama-3.1-8B model (Dubey *et al.*, 2024) is a prominent example of this design and has been adopted as the backbone in recent KG-constrained decoding work (Luo *et al.*, 2025).

Text production in these models is fundamentally autoregressive: given a vocabulary $\mathcal{V}$, an input x , and prior tokens $y_{<t}$, the model computes $P(y_t|y_{<t}, x)$ over the full vocabulary. Since this distribution is always defined over every token, any token has a positive, non-zero chance of being generated at any step. Decoding strategies such as beam search and nucleus sampling (Holtzman *et al.*, 2020) determine how a token is chosen from this distribution.

This autoregressive nature creates fluent, coherent text, but since each step relies on prior outputs without an external verifier, it also directly causes hallucination.

LLMs are pretrained on vast unlabelled text using next-token prediction. The GPT series (Brown *et al.* 2020; OpenAI 2024) and the Llama series (Touvron, Martin, and Stone 2023; Dubey *et al.* 2024) show that scaling this simple task leads to powerful downstream capabilities. During pretraining, the model internalizes patterns and facts from its training data, but this memory is fundamentally static and statistical. The model lacks a way to interact with dynamic external facts or ensure that its outputs reflect verified truths.

2.1.1 Definition and Structural Causes

Hallucination in natural language generation refers to the production of content that is fluent and internally coherent yet factually unsupported by or in contradiction with verifiable sources (Ji *et al.*, 2023). Huang *et al.* (2025) refine this into *intrinsic hallucination* (output contradicts provided source context) and *extrinsic hallucination* (output contains claims unverifiable against any source). Both types appear at non-trivial rates: Ji *et al.* (2023) survey over 30 works finding hallucination rates from 15% to over 60%, and Luo *et al.* (2025) report that retrieval-augmented reasoning approaches produce hallucinated reasoning steps in approximately one-third of cases even with direct access to a knowledge graph.

Hallucination is generally attributed to a combination of data, training, and inference-time factors (Huang *et al.*, 2025). This work is concerned with the last of these: the autoregressive decoding mechanism itself makes hallucination structurally possible, independent of data quality or training procedure. Three properties of this mechanism are relevant. First, the generation distribution is based entirely on learned parameters, with no component that verifies tokens against an external source. Second, errors propagate forward: a hallucinated token at step t becomes conditioning context for step $t+1$, producing fluent continuations of an incorrect chain. Third, the model imposes no hard constraint on which tokens are eligible at each step, so factually invalid continuations remain within the support of the output distribution (Brown *et al.*, 2020).

2.1.2 The Scaling Paradox

A common argument is that larger models should eventually achieve factual reliability. Evidence does not support this. Lin, Hilton, and Evans (2022) introduce TruthfulQA and find that larger models do not outperform smaller ones in truthfulness and in some categories show *inverse scaling*, reproducing widespread incorrect beliefs more effectively. Perez *et al.* (2023) show that RLHF-trained models generate confident-sounding responses aligned with apparent user expectations rather than verifiable facts. Kandpal *et al.* (2023) demonstrate that larger models improve on frequently-seen entity combinations but not on compositional questions requiring unseen fact combinations, precisely the structure of multi-hop KG questions. Mallen *et al.* (2023) confirm significant hallucination rates for GPT-4 on tail-entity questions regardless of model size. Together, these findings establish hallucination as a structural feature of the autoregressive mechanism, not a scaling issue.

2.2 Chain-of-Thought and Its Limits

2.2.1 Chain-of-Thought Prompting

Chain-of-thought (CoT) prompting (Bosma *et al.*, 2022) is the dominant paradigm for eliciting multi-step reasoning from LLMs. By including worked examples that show intermediate reasoning steps, the model learns to decompose complex problems into simpler sub-problems. Bosma *et al.* (2022) demonstrate accuracy improvements on arithmetic, commonsense, and multi-hop QA benchmarks, with the capability emerging at approximately 100 billion parameters. Zero-shot chain-of-thought (Kojima *et al.*, 2022) achieves similar benefits by adding “Let’s think step by step” to the prompt.

Despite these improvements, CoT has a fundamental structural limitation: each intermediate step is generated by the same autoregressive mechanism. An incorrect step propagates forward as conditioning context, and no mechanism verifies that any step corresponds to a fact in an external source. A CoT chain can be entirely fluent and internally coherent while containing factually fabricated steps.

2.2.2 Self-Consistency and Tree-of-Thought

Self-consistency (Wang *et al.*, 2023) samples several independent CoT chains and chooses the final answer by majority vote, reducing sensitivity to individual hallucinated chains. However, a majority of independently hallucinated chains can still produce an incorrect answer, especially on deep multi-hop questions. Tree-of-Thought (Cao *et al.*, 2023) treats generation as search through a tree of partial reasoning sequences, using the LLM as both generator and evaluator. The key limitation persists: pruning decisions rely on the model’s own parameters without external verification of factual accuracy.

2.2.3 Why Prompting Cannot Guarantee Faithfulness

All prompting-based methods (few-shot, CoT, self-consistency, tree-of-thought) modify the *input* to the LLM without modifying the *generation mechanism*. Since decoding happens over the full vocabulary, each token has a strictly positive probability at every step. True structural faithfulness requires that invalid tokens receive exactly zero probability. This can only be achieved by directly changing the decoding algorithm (Geng *et al.*, 2023; Willard and Louf, 2023).

2.3 Knowledge Graphs and Knowledge Graph Question Answering

2.3.1 Knowledge Graphs

A knowledge graph is a structured database of verified facts, represented as directed, labelled triplets (e, r, e') where $e, e' \in \mathcal{E}$ are entities and $r \in \mathcal{R}$ is a relation type. Figure 2.1 illustrates a simplified fragment of such a structure. Freebase (Bollacker *et al.*, 2008), the KG underlying the evaluation benchmarks used in this work, encodes tens of millions of triplets across geography, history, entertainment, and science (Yih *et al.*, 2016; Talmor and Berant, 2018).

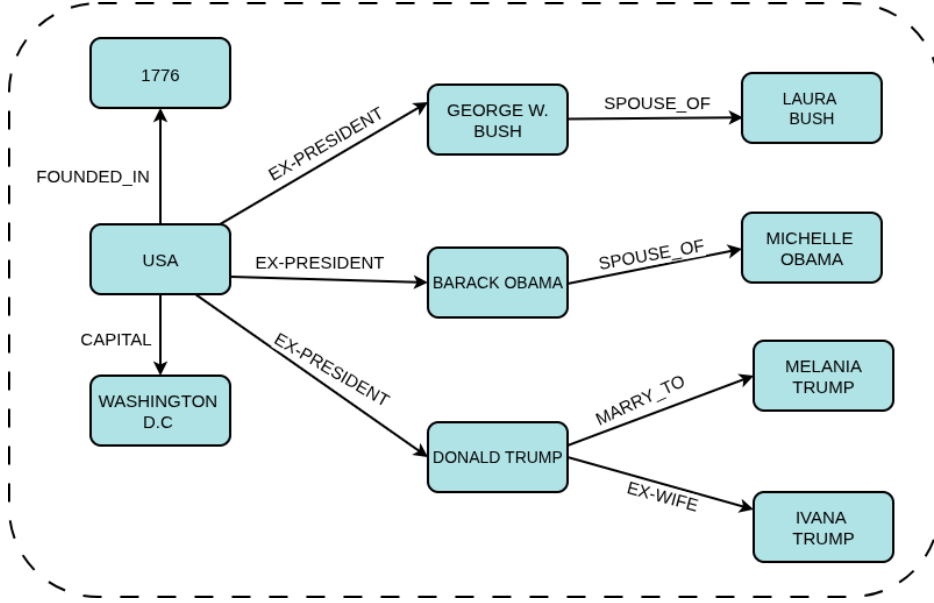


Figure 2.1 Knowledge Graph Fragment

The critical distinction between a KG and an unstructured text corpus is that every triplet has been explicitly verified and is machine-readable in deterministic form. This makes KGs suitable as sources for hard constraints on generation: a candidate token can be verified via exact lookup rather than probabilistic semantic matching. A multi-hop reasoning path of length L is a sequence $(e_0, r_1, e_1, \dots, r_L, e_L)$ in which each consecutive triplet (e_{i-1}, r_i, e_i) is a verified edge of $\mathcal{G}$ (Lan *et al.*, 2022).

2.3.2 Knowledge Graph Question Answering

KGQA requires a system to identify the answer entity e_L to a natural language question q by reasoning through a chain of verified KG triplets. Formally: given question q with question entities $\mathcal{E}_q \subseteq \mathcal{E}$ identified by entity linking, identify the terminal entity of the correct reasoning path in $\mathcal{G}$ (Lan *et al.*, 2022). Early approaches relied on semantic parsing, information retrieval over KG subgraphs, or reinforcement-learning-based path traversal (Das *et al.*, 2018). More recent approaches use LLMs as retrievers, agents iteratively querying the KG and revising partial answers (Mavromatis and Karypis, 2022), generators conditioned on retrieved subgraph context (Jiang *et al.*, 2023a; Jin *et al.*, 2024; Luo *et al.*, 2024), unified retrieval-and-reasoning pipelines (Jiang *et al.*, 2022), or KG-infused pretrained representations (Wang *et al.*, 2021). The move toward LLM-based KGQA has improved fluency but brought back hallucination risks.

2.3.3 Evaluation Benchmarks

WebQSP (Yih *et al.*, 2016) contains 4,737 questions on Freebase with reasoning depths of one to two hops, evaluated using Hits@1 and F1. ComplexWebQuestions (CWQ; Talmor and Berant 2018) builds on WebQSP through conjunction, composition, and comparative operations, resulting in about 34,000 questions requiring up to four hops. The two benchmarks stress different aspects of path selection: WebQSP questions reach the answer in one or two hops, so the valid path set stays small; CWQ questions reach up to four hops, so the valid paths are longer and correct selection depends on sustaining the right trajectory across more steps, not on filtering a larger candidate set.

2.3.4 Multi-Hop Complexity

For an entity with average out-degree d in the KG, the number of valid paths of length L from the question entities increases as $|\mathcal{E}_q| \times d^L$. For well-connected Freebase entities where $d \approx 20$ and a question may have up to three question entities, a three-hop question can generate up to 24,000 structurally valid paths, but only one leads to the correct answer (He *et al.*, 2021; Lan *et al.*, 2022). At each step, the model must choose the correct next token from a vast array of valid yet incorrect options.

2.4 Constrained Decoding for Faithful Text Generation

Constrained decoding restricts generation to a predefined valid set at each step by directly intervening in the decoding mechanism. Unlike prompting, which changes the input to the LLM, constrained decoding alters the token selection process itself.

2.4.1 Format-Constrained Decoding

Logit masking (Geng *et al.*, 2023; Willard and Louf, 2023) computes a binary mask over the vocabulary at each generation step, setting the log-probability of invalid tokens to $-\infty$ before the softmax. Geng *et al.* (2023) constrain generation to strings conforming to a context-free grammar via an Earley parser oracle. Willard and Louf (2023) reduce the computational cost with an efficient finite-state machine (FSM) formulation, precomputing an index from automaton states to valid vocabulary subsets for amortised $O(1)$ lookup. Additional methods include Synchromesh (Poesia *et al.*, 2022) for program synthesis, PICARD (Scholak, Schucher,

and Bahdanau, 2021) for schema-constrained SQL parsing, NeuroLogic Decoding (Lu *et al.*, 2021) for propositional logic constraints, and GENRE (De Cao *et al.*, 2021) for trie-constrained entity retrieval. The shared limitation of all these approaches is that they enforce output *format* rather than *factual content*. The extension from format constraints to KG path constraints requires a constraint oracle that encodes the relational structure of the knowledge graph itself.

2.4.2 Knowledge Graph-Constrained Decoding

Applying constrained decoding to knowledge graph faithfulness uses the KG itself as the constraint oracle. At each generation step, only tokens corresponding to valid continuations of a KG reasoning path are allowed, making hallucination of KG facts structurally impossible.

Graph-Constrained Reasoning (GCR) (Luo *et al.*, 2025) builds a KG-Trie (a prefix tree of all KG paths within L hops of the question entities, created by BFS before generation) and uses it as the constraint oracle during beam-search decoding. GCR achieves 92.6% Hits@1 on WebQSP and 75.8% on CWQ with 100% structural faithfulness, using a Llama-3.1-8B backbone. The trie is created once and remains unchanged: the valid token set is fixed regardless of what has been generated.

Decoding on Graphs (DoG) (Li *et al.*, 2025) presents step-wise trie expansion as a partial solution to GCR’s static oracle. After locking in an entity, the constraint trie is expanded with all KG neighbors of that entity, making the oracle topologically dynamic. DoG achieves 90.99% Hits@1 on WebQSP and 76.08% on CWQ with 100% structural faithfulness. However, the expansion rule includes all KG neighbors regardless of relevance to the question, and step-wise trie construction compromises the efficiency of precomputed structures.

RouterKGQA (Yuan *et al.*, 2026) routes between a specialized KG-reasoning model and a general LLM based on question complexity. It uses SBERT embeddings (`nomic-embed-text-v1`) to filter candidate answers after generation. RouterKGQA’s semantic scoring operates at the answer selection level rather than at the token constraint level: paths are created first and filtered afterward.

2.4.3 Beam Search and Trie-Based Constraints

Beam search (Lowerre, 1976; Holtzman *et al.*, 2020) keeps the k most promising partial hypotheses at each step, reducing early-error multiplication. A prefix trie (Fredkin, 1960; Willard and Louf, 2023) yields the valid next-token subset in $O(1)$ time. GCR’s KG-Trie (Luo *et al.*, 2025) serializes verified multi-hop paths from the KG into string format mapped to the LLM’s token vocabulary. When beam search is merged with a KG-Trie oracle, the decoding process becomes a strictly bounded traversal of the prefix tree: at each autoregressive step, the current beam hypotheses map to specific trie states, and a mathematical mask forces the probability of any token not present as a child node to zero. The beam search then advances strictly down permissible branches.

2.5 Retrieval and Soft Grounding

Retrieval-augmented generation (RAG) (Lewis *et al.*, 2020) is the leading approach for grounding LLM outputs in external knowledge. A dense retriever selects relevant passages based on embedding similarity, and a reader LLM generates the answer from the retrieved context. Fusion-in-Decoder (Izacard and Grave, 2021) enhances this by encoding multiple passages separately and merging them in the decoder. Self-RAG (Asai *et al.*, 2024) adds reflection tokens for when to retrieve and whether passages support claims. FLARE (Jiang *et al.*, 2023b) triggers retrieval when next-token confidence drops. IRCoT (Trivedi *et al.*, 2023) interleaves retrieval with CoT steps, improving multi-hop accuracy. GraphRAG (Edge *et al.*, 2024) builds community-structured KGs from document collections for complex relational queries.

In the KGQA context, several approaches enhance RAG with structured context. He *et al.* (2021) retrieve KG subgraphs with attention-guided traversal. Yasunaga *et al.* (2021) combine text passages and KG neighborhoods through graph neural networks. Shi *et al.* (2021) retrieve KG paths as natural language context strings. Sentence-BERT (Reimers and Gurevych, 2019) encoders produce dense vector representations where cosine similarity proxies semantic alignment, applied in these systems to rank candidate KG paths as soft retrieval signals (Shi *et al.*, 2021; Saxena, Chakrabarti, and Talukdar, 2022).

2.5.1 The Structural Limitation of Soft Grounding

Despite their empirical improvements, all RAG approaches share a structural limitation: retrieved context is a soft influence on generation. As shown in Figure 2.2, the retriever determines what the LLM sees; it does not determine what the LLM is permitted to generate. Because the generation mechanism remains unchanged, there exists a nonzero probability of generating any token at any step, including tokens that correspond to no verified KG fact. Structural faithfulness with probability one cannot be achieved under any architecture that modifies only the input context without modifying the token selection mechanism (Luo *et al.*, 2025; Geng *et al.*, 2023).

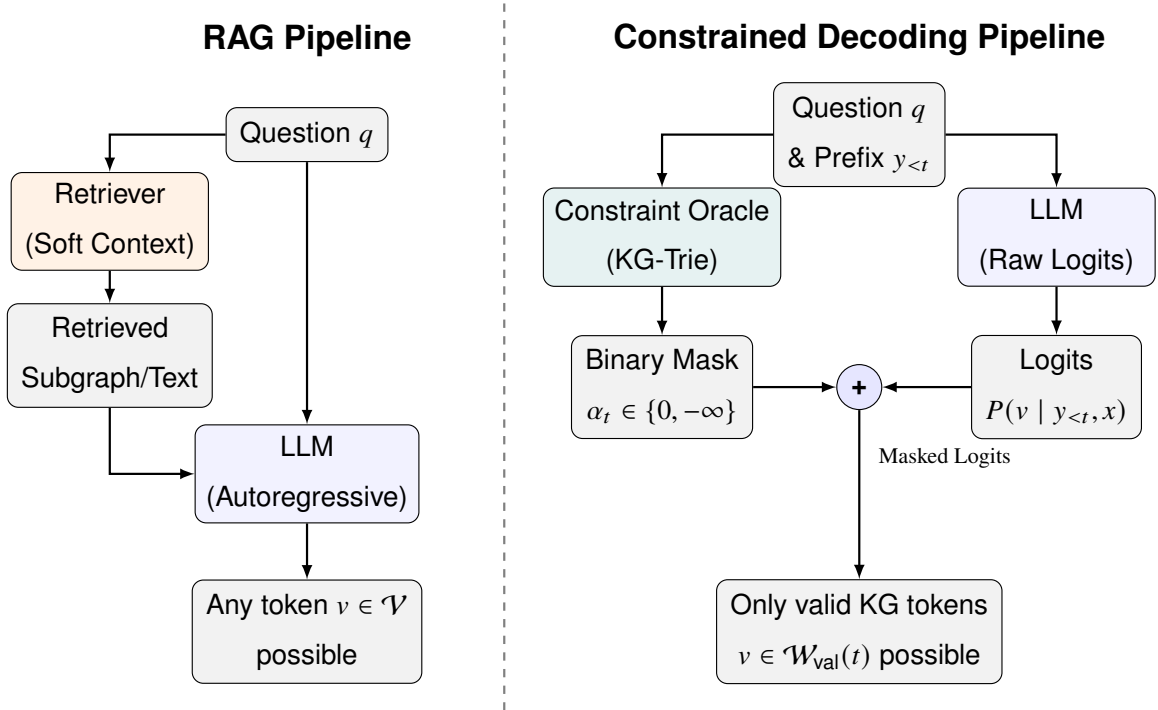


Figure 2.2 Standard RAG vs. KG-Constrained Decoding

2.6 Unresolved Gaps

The review shows that two different research paths, constrained decoding for structural faithfulness and semantic similarity for relevance-guided reasoning, have developed mainly in parallel without being combined at the constraint-oracle level. Constrained decoding methods (Luo *et al.*, 2025; Li *et al.*, 2025) achieve structural faithfulness but create their constraint oracles without considering question semantics or the context of generation. This leads to valid token sets that include many irrelevant paths. Semantic similarity methods (Shi

et al., 2021; Saxena, Chakrabarti, and Talukdar, 2022; Yuan *et al.*, 2026) improve relevance but act as soft signals that affect retrieval or selection after the fact, without changing the generation process.

Three specific gaps emerge from this review.

Gap 1: The Permissiveness Problem. Current KG-constrained decoding frameworks build their constraint oracles based on graph structure and question entities only. They do not consider question semantics or the current generation state. In the case of multi-hop questions, this results in valid token sets with thousands of structurally correct but semantically irrelevant paths at each generation step. The LLM has to sort through these paths using its own knowledge, which reintroduces the risk of hallucination that constrained decoding aimed to eliminate (Luo *et al.*, 2025; Li *et al.*, 2025).

Gap 2: The Static-Dynamic Efficiency Tradeoff. GCR’s precomputed static trie is efficient in computing but lacks semantic awareness and remains the same at every generation step. DoG’s step-wise topological expansion adds some dynamism but requires reconstructing the trie at every step and does not include semantic filtering (Li *et al.*, 2025; Willard and Louf, 2023). No current framework achieves semantic responsiveness, where the valid set narrows progressively as reasoning moves in a specific direction, while also maintaining the efficiency of precomputed constraint structures.

Gap 3: The Absence of a Constraint Quality Metric. No existing work offers a metric that measures constraint permissiveness separately from the accuracy of downstream answers. Without this metric, it is not possible to track progress on reducing irrelevant paths in the valid set or to compare frameworks based on constraint quality without factoring in model quality.

Table 2.1 summarizes the key works reviewed in this chapter, categorizing them by their approach, whether they provide structural faithfulness guarantees, and whether they incorporate semantic conditioning into their constraint mechanisms.

Table 2.1 Summary of Key Related Works

Work	Approach	Structural Faithfulness	Semantic Conditioning
Luo <i>et al.</i> (2025)	Static KG-Trie decoding	Yes (100%)	No
Li <i>et al.</i> (2025)	Step-wise topological expansion	Yes (100%)	No
Yuan <i>et al.</i> (2026)	Routing + post-hoc filtering	Probabilistic	Yes (answer-level)
Luo <i>et al.</i> (2024)	Reasoning path planning	No	Partial
Jiang <i>et al.</i> (2023a)	LLM over structured data	No	Partial
Jin <i>et al.</i> (2024)	Graph-augmented CoT	No	Partial
Jiang <i>et al.</i> (2022)	Unified retrieval and reasoning	No	Partial
Wang <i>et al.</i> (2021)	KG-pretrained language model	No	Partial
Mavromatis and Karypis (2022)	Iterative reasoning revision	No	Partial
He <i>et al.</i> (2021)	KG subgraph attention	No	Partial
Yasunaga <i>et al.</i> (2021)	Text-KG graph neural network	No	Partial
Das <i>et al.</i> (2018)	RL-based KG path traversal	No	Soft
Saxena, Chakrabarti, and Talukdar (2022)	Embedding-based path selection	No	Soft
Shi <i>et al.</i> (2021)	Path retrieval by similarity	No	Soft (retrieval)
Asai <i>et al.</i> (2024)	Retrieval with reflection	No	Soft
Edge <i>et al.</i> (2024)	Community-structured retrieval	No	Soft

Work	Approach	Structural Faithfulness	Semantic Conditioning
Geng <i>et al.</i> (2023)	Grammar-constrained decoding	Format only	No
Willard and Louf (2023)	FSM-indexed constrained decoding	Format only	No
De Cao <i>et al.</i> (2021)	Entity trie-constrained decoding	Entity names only	No
Scholak, Schucher, and Bahdanau (2021)	Schema-constrained SQL parsing	Schema only	No
Lu <i>et al.</i> (2021)	Logic-constrained decoding	Format only	No

These three gaps collectively identify an open problem at the intersection of the two research trajectories reviewed in this chapter: integrating semantic relevance scoring directly into the constraint oracle, conditioning the valid token set jointly on the knowledge graph, the question semantics, and the evolving generation state.

CHAPTER 3

METHODOLOGY

3.1 Overview

This chapter formalises the permissiveness problem identified in Chapter 2 and presents the Dynamic Context-Aware Trie (DCA-Trie) framework as a solution. The upgraded oracle specification and the Semantic Irrelevance Ratio (SIR) are defined first, with SIR serving as a diagnostic metric that measures constraint quality independently of answer accuracy. The chapter then traces the research evolution from an initial cosine-similarity scorer through a decomposed product score to the final symbolic TypeOracle, explaining the rationale behind each design choice. The static pre-filtering design of v1 and the step-wise dynamic expansion design of v2 are described using the TypeOracle mechanism. The chapter closes with the baseline configurations, evaluation protocol, and scope boundaries that govern the experimental work in Chapter 4.

3.2 Formal Problem Specification

3.2.1 Core Limitation of Existing Frameworks

As established in Chapter 2, current graph-constrained frameworks (notably GCR (Luo *et al.*, 2025) and DoG (Li *et al.*, 2025)) rely on a deterministic oracle that looks only at the fixed graph $\mathcal{G}$ and the starting entities $\mathcal{E}_q$. Regardless of whether the system builds this oracle before decoding or expands it on the fly, the set of valid tokens at any step t remains locked to structural logic:

$$\mathcal{W}_{\text{val}}^{\text{GCR}}(t) = f(\mathcal{G}, \mathcal{E}_q) \quad \forall t \quad (3.1)$$

Equation (3.1) captures this static constraint model. By ignoring both the semantic intent of the question q and the reasoning trajectory already captured in the prefix $y_{<t}$, this formulation treats every structurally reachable path as equally plausible. In dense, multi-hop Freebase environments, this causes substantial search space inflation. On WebQSP, the oracle admits an average of 2,553 paths per question (Chapter 4, Table 4.3), even though only a single path

logically answers the query.

Two separate requirements are in tension here. Structural validity ensures the LLM only generates tokens that correspond to real graph edges; current models do this well. Contextual relevance is the open problem: ensuring the LLM only generates tokens that logically connect to the specific question. Current models do not enforce this. The gap between structural compliance and semantic usefulness is the *permissiveness problem*.

3.2.2 The Upgraded Oracle Specification

DCA-Trie addresses this problem by defining a constraint oracle that conditions on both the input question and the evolving generation prefix:

$$\mathcal{W}_{\text{val}}^{\text{DCA}}(t) = f(\mathcal{G}, \mathcal{E}_q, q, y_{<t}) \quad (3.2)$$

where q is the natural language query and $y_{<t} = (y_1, \dots, y_{t-1})$ is the generated prefix at step t . The upgraded oracle must satisfy three conditions aligned with the project objectives in Chapter 1:

1. **Structural Faithfulness (Condition F):** Every candidate token $v \in \mathcal{W}_{\text{val}}^{\text{DCA}}(t)$ must correspond to a valid prefix of a path $p \in \mathcal{G}$ at every step t . This preserves the 100% structural faithfulness guarantee achieved by GCR and DoG and prevents ungrounded tokens from entering the valid set.
2. **Semantic Relevance Reduction (Condition R):** The Semantic Irrelevance Ratio (SIR) under DCA-Trie must be lower than the corresponding SIR under GCR when averaged over the evaluation corpus. Using SIR as defined in Section 3.4:

$$\overline{\text{SIR}}(\mathcal{W}^{\text{DCA}}) < \overline{\text{SIR}}(\mathcal{W}^{\text{GCR}}) \quad (3.3)$$

3. **Recall Preservation (Condition P):** Semantic pruning must not remove gold paths excessively. Operationally, the false negative rate (FNR) on the WebQSP validation split must remain below 5%, as formalised in Equation (3.4):

$$\text{FNR} = \frac{|\{q \in Q_{\text{val}} : p_q^* \notin \mathcal{W}_{\text{val}}^{\text{DCA}}(t)\}|}{|Q_{\text{val}}|} < 0.05 \quad (3.4)$$

where p_q^* is the gold path for question q and Q_{val} is the held-out 100-question validation set.

These three conditions are not guaranteed to be jointly satisfiable. Tightening the oracle to satisfy Condition R (reduce irrelevance) can directly threaten Condition P (remove gold paths) when the semantic signal mismatches the lexical form of correct answers. This tension is the central axis of the Chapter 4 results.

3.3 System Architecture Overview

DCA-Trie focuses on the constraint oracle layer, so its architecture directly builds on the baseline pipeline from Graph-Constrained Reasoning (GCR) (Luo *et al.*, 2025). This reuse helps isolate the experimental effects of the proposed oracle. The four layers work as follows:

1. **Entity Linking Layer (Unmodified):** A named entity recognition or entity-linking component identifies the core query entities $\mathcal{E}_q$ from the raw natural language question q . These entities dictate where downstream graph exploration begins.
2. **Constraint Oracle Layer (DCA-Trie Contribution):** This layer controls which tokens the LLM can generate. In GCR, this layer creates a KG-Trie that includes structurally reachable paths within L hops of the starting entities. DCA-Trie modifies this layer by applying symbolic ontology checks through a TypeOracle. Through either static pre-filtering (v1) or dynamic step-wise expansion (v2), the oracle ensures that admitted paths remain graph-valid and type-consistent.
3. **Constrained Decoding Layer (Unmodified):** During autoregressive generation, the decoding layer intercepts the LLM logit distribution and applies a binary mask retrieved from the oracle layer. The LLM is therefore restricted to tokens that are currently valid under the trie.
4. **Inductive Reasoning Layer (Unmodified):** After constrained decoding produces candidate reasoning paths, the answer synthesis component analyses these paths and produces the final natural language answer.

Figure 3.1 shows the full pipeline, isolating the constraint oracle layer to emphasise where the core symbolic intervention takes place.

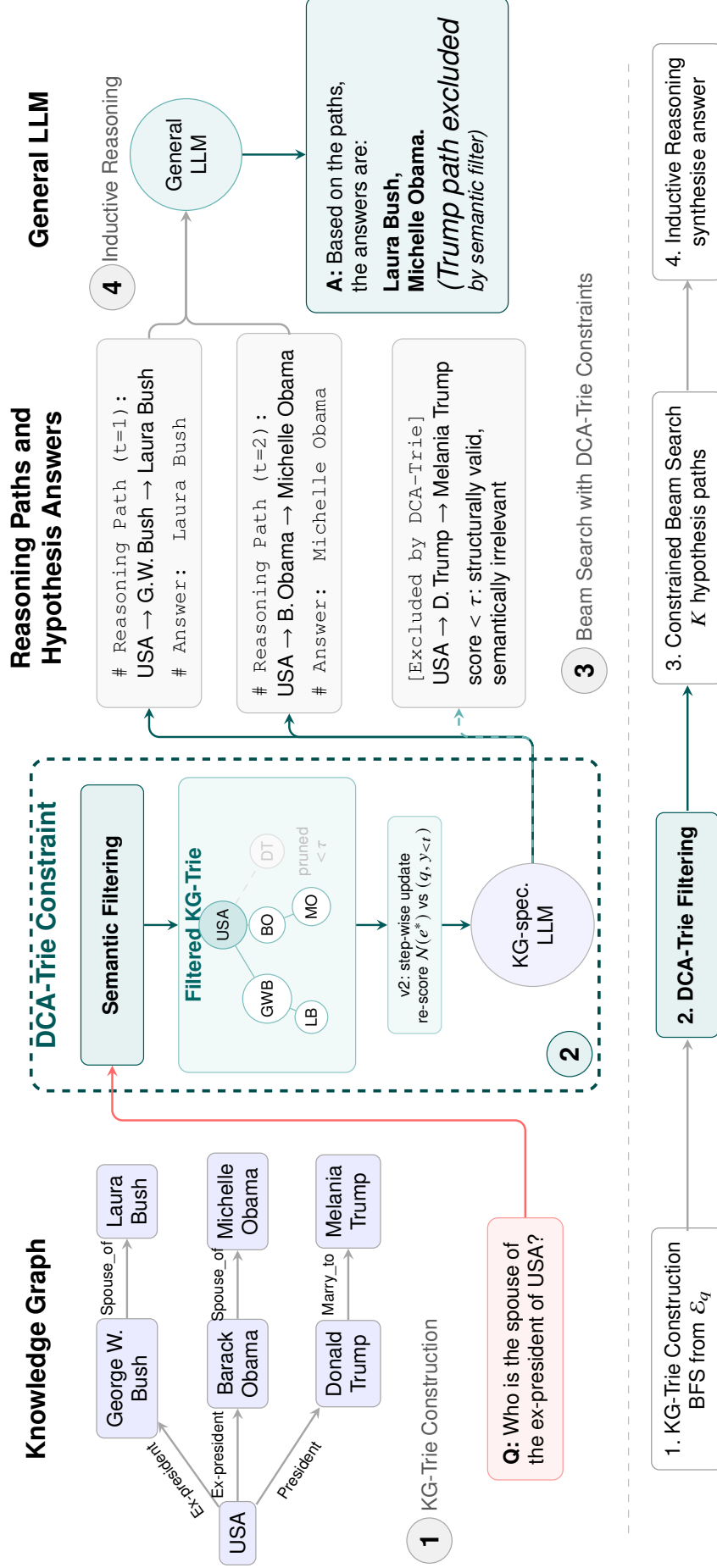


Figure 3.1 DCA-Trie System Architecture.

3.4 Semantic Irrelevance Ratio

3.4.1 Standard Metric Deficiencies

Current performance markers for knowledge graph decoding, such as Hits@1, F1, and structural faithfulness ratios, have a significant drawback: they focus on output instead of the process. While these metrics show whether a framework found the correct answer and stayed within the graph, they do not reveal how the framework reached that answer. A model may solve the correct gold path, but it could wander through a large number of irrelevant paths that waste resources.

Confirming accuracy is not enough; the constraint mechanism itself must be measured. To assess how permissive the oracle is, independently of final answer accuracy, this thesis presents a diagnostic metric called the Semantic Irrelevance Ratio (SIR).

3.4.2 Formal Definition of SIR

Suppose $\mathcal{P}(q, t)$ represents all paths that the constraint oracle allows at decoding step t for question q . To calculate SIR, the system must first define irrelevance. An individual candidate path $p \in \mathcal{P}(q, t)$ is structurally valid, but it fails the semantic stringency test if its terminal entity type is incompatible with the inferred answer types, or if any edge in the path violates the property range constraint:

$$\text{irrelevant}(p, q) = \mathbf{1}[\neg \text{type_gate}(p, q) \vee \exists (e, r, e') \in p : \neg \text{range_gate}(r, e')] \quad (3.5)$$

Equation (3.5) defines the binary irrelevance indicator. Here, `type_gate` checks whether the terminal entity’s type matches the expected answer types inferred from the question, and `range_gate` checks whether each edge respects the ontology’s declared property ranges. Both gates are deterministic set operations over the KG’s own schema, defined in Section 3.5.2.

The step-level SIR for question q at decoding step t is then defined as:

$$\text{SIR}(q, t) = \frac{\sum_{p \in \mathcal{P}(q, t)} \text{irrelevant}(p, q)}{|\mathcal{P}(q, t)|} \quad (3.6)$$

As shown in Equation (3.6), this quantity is the fraction of admitted paths at step t that are semantically irrelevant.

Corpus-level SIR is computed by averaging over all questions and all valid decoding steps up to depth L :

$$\overline{\text{SIR}} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{L_q} \sum_{t=1}^{L_q} \text{SIR}(q, t) \quad (3.7)$$

Equation (3.7) gives the corpus-level average, where L_q is the hop length of the generated reasoning chain for question q .

3.4.3 Properties and Interpretation

By definition, $\overline{\text{SIR}} \in [0, 1]$. A value near 1 indicates that most admitted paths are irrelevant, so the model still faces a large and noisy search space. A value near 0 indicates that the constraint set is tightly aligned with question semantics. Prior constrained frameworks such as GCR and DoG tend to show high SIR, especially at larger hop depths where path counts increase rapidly. DCA-Trie targets lower SIR by filtering semantically weak paths. The structurally valid ones stay. Chapter 4 compares SIR across WebQSP and CWQ, whose differing hop depths give an indication of how this effect scales with reasoning complexity.

3.5 Symbolic Relevance Scoring Mechanism

3.5.1 Design Evolution

The semantic relevance mechanism went through three designs before arriving at the final symbolic TypeOracle. The initial cosine-similarity scorer (Approach 1) used sentence-transformer embeddings to score each candidate path against the question. It suffered from semantic collapse (a single score cannot distinguish type-match from topical overlap), encoder cost (thousands of forward passes per question), threshold sensitivity ($\tau = 0.25$ produced empty path sets for 84% of WebQSP queries), and non-determinism from floating-point arithmetic. The decomposed product score (Approach 2) split the cosine score into relational, entity-type, and trajectory components. The entity-type component was already a hard binary gate rather than a continuous score, an early step toward the fully symbolic design, but the relational and trajectory components still required encoder forward passes. Approach 2 improved interpretability but retained both the encoder dependency and the threshold. The final TypeOracle (Approach 3) eliminates the encoder and the threshold entirely, using only $O(1)$ set lookups over the KG’s own ontology metadata.

Table 3.1 summarises the properties of all three designs. Detailed design documents for Approaches 1 and 2 are provided in Appendix A.

Table 3.1 Comparison of Oracle Designs

Property	Approach 1 Cosine	Approach 2 Decomposed	Approach 3 TypeOracle
Encoder needed	Yes	Yes	No
Threshold τ	Yes	Yes	No
Type checking	Implicit	Hard gate	Ontology-based
Range checking	None	None	Ontology-based
Per-path cost	Forward pass	Forward pass	O(1) set lookup
Deterministic	No	No	Yes
Status	Abandoned	Abandoned	Final

3.5.2 TypeOracle Architecture

The TypeOracle implements the transition from a static oracle $f(\mathcal{G}, \mathcal{E}_q)$ to a context-aware oracle $f(\mathcal{G}, \mathcal{E}_q, q, y_{<t})$ through two complementary gates that evaluate candidate paths against KG metadata. Figure 3.2 illustrates the admission process.

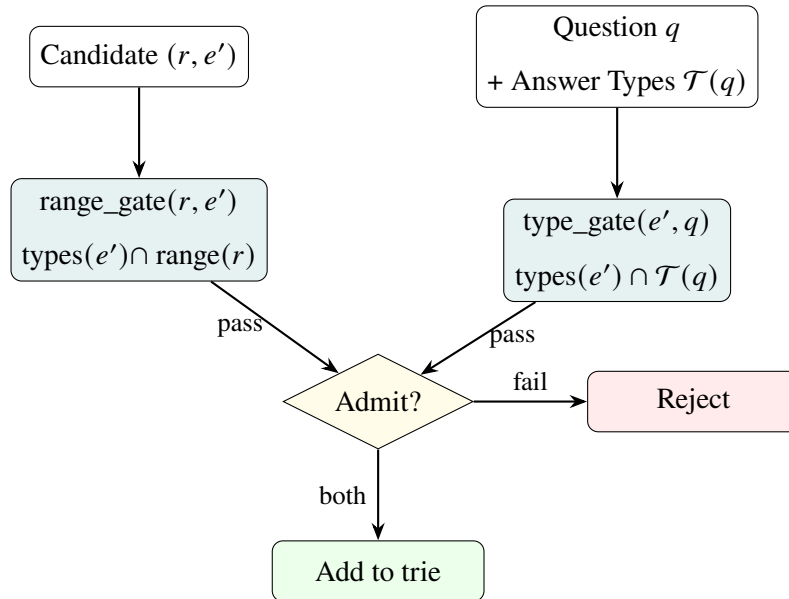


Figure 3.2 TypeOracle Admission Process

The oracle consists of two complementary gates:

1. **Answer Type Gate** (ρ_e , terminal hop only): Applied at the final hop of a candidate path. It checks whether the candidate answer entity’s type matches what the question asks for. For example, a question beginning with “who” infers person-type entities as valid answers.
2. **Property Range Gate** (ρ_r , every hop): Applied at every edge in the path. It checks whether the tail entity’s type is compatible with the relation’s declared range in the ontology. For example, if a relation expects a location as its range, then only entities typed as locations are admitted when type information is available.

The combined admission check for a candidate edge (r, e') at hop h is:

$$\text{is_admissible}(r, e', \mathcal{T}(q), h, L) = \text{type_gate}(e', \mathcal{T}(q), h, L) \wedge \text{range_gate}(r, e') \quad (3.8)$$

where $\mathcal{T}(q)$ is the set of expected answer types inferred from the question, h is the current hop, and L is the maximum hop depth. A candidate edge is admitted only if it passes all active gates.

3.5.3 Ontology Schema and Threshold-Free Design

The TypeOracle is constructed from the knowledge graph’s own schema triples. Freebase encodes entity types through `common.topic.notable_types`, property domains through `rdf-schema#domain`, and property ranges through `rdf-schema#range`. From these, the oracle builds two lookup structures: `_entity_types` (entity-to-types) and `_schema` (relation-to-range). Question type inference uses lightweight pattern matching (“who” $\rightarrow$ person, “where” $\rightarrow$ location, etc.). If no pattern matches, the type gate admits candidates conservatively.

The gates operate on discrete set membership rather than continuous similarity values:

$$\text{type_gate}(e', \mathcal{T}(q), h, L) = \begin{cases} \text{True} & h < L \\ \text{True} & \mathcal{T}(q) = \emptyset \\ \text{True} & \text{types}(e') = \emptyset \\ \mathbf{1}[\text{types}(e') \cap \mathcal{T}(q) \neq \emptyset] & \text{otherwise} \end{cases} \quad (3.9)$$

$$\text{range_gate}(r, e') = \begin{cases} \text{True} & r \notin \mathcal{R} \\ \text{True} & \text{range}(r) = \emptyset \\ \text{True} & \text{types}(e') = \emptyset \\ \mathbf{1}[\text{types}(e') \cap \text{range}(r) \neq \emptyset] & \text{otherwise} \end{cases} \quad (3.10)$$

Both gates default to True when schema metadata is absent. This ensures structural faithfulness is never compromised and the TypeOracle never produces empty path sets. The design is threshold-free: no continuous similarity score requires tuning. The computational properties are summarised in Table 3.3.

3.5.4 Computational Properties

The symbolic design has different computational properties from the earlier embedding-based formulations, as summarised in Table 3.3.

Table 3.3 Computational Properties: Embedding vs. Symbolic

Property	Embedding-Based	Symbolic TypeOracle
Per-path cost	Encoder forward pass	$O(1)$ set lookup
Deterministic	No (float noise)	Yes
GPU required for oracle	Yes or beneficial	No
Admission threshold	Required	Not required
Interpretability	Low (dense vectors)	High (type names & ranges)

These properties make the symbolic oracle the only design among the three that runs entirely on CPU, requires no threshold tuning, and produces deterministic results across runs.

3.6 DCA-Trie v1

3.6.1 Design Rationale

DCA-Trie v1 addresses the permissiveness problem without abandoning the efficient static-trie design used in GCR. The key idea is to apply symbolic filtering once, before decoding begins, using the KG’s ontology schema and the inferred answer type of the question. This is useful because many paths are structurally reachable but type-incompatible with the question’s intent. Filtering paths against the ontology during preprocessing removes weak candidates early and reduces the search space before autoregressive generation starts.

Figure 3.3 summarises this offline filtering workflow.

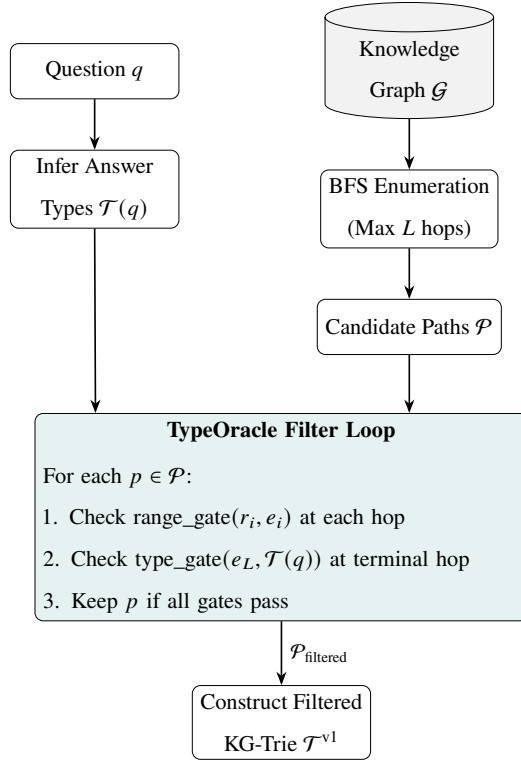


Figure 3.3 Static Symbolic Filtering Pipeline (DCA-Trie v1)

3.6.2 Algorithm for DCA-Trie v1

Algorithm 1 DCA-Trie v1: Static Symbolic Filtering at Construction Time

Require: Knowledge graph $\mathcal{G}$; question entities $\mathcal{E}_q$; question q ; max hop depth L ; TypeOracle O

Ensure: Semantically filtered KG-Trie $\mathcal{T}^{v1}$

```

1: Infer answer types:  $\mathcal{T}(q) \leftarrow O.infer\_answer\_types(q)$ 
2: Enumerate paths:  $\mathcal{P} \leftarrow \text{BFS}(\mathcal{G}, \mathcal{E}_q, L)$  ▷ All paths within  $L$  hops, as in GCR
3: Filter through TypeOracle:  $\mathcal{P}_{\text{filtered}} \leftarrow \emptyset$ 
4: for each path  $p = (e_0, r_1, e_1, \dots, r_L, e_L) \in \mathcal{P}$  do
5:   admit  $\leftarrow$  True
6:   for each hop  $(r_i, e_i)$  in  $p$  do
7:     if  $\neg O.range\_gate(r_i, e_i)$  then
8:       admit  $\leftarrow$  False; break
9:     end if
10:  end for
11:  if admit  $\wedge \neg O.type\_gate(e_L, \mathcal{T}(q), L, L)$  then
12:    admit  $\leftarrow$  False
13:  end if
14:  if admit then
15:     $\mathcal{P}_{\text{filtered}} \leftarrow \mathcal{P}_{\text{filtered}} \cup \{p\}$ 
16:  end if
17: end for
18: Construct filtered trie:  $\mathcal{T}^{v1} \leftarrow \text{BUILDTRIE}(\mathcal{P}_{\text{filtered}})$ 
19: return  $\mathcal{T}^{v1}$ 

```

The process unfolds in five stages:

1. **Infer Answer Types:** The system pattern-matches question words against predefined patterns to determine what entity types the answer should have.
2. **Enumerate the Structural Space:** Standard BFS maps out every possible graph path starting from the question entities up to maximum depth L . This matches the baseline GCR approach.
3. **Apply TypeOracle Filtering:** For each enumerated path, the oracle checks whether every edge respects the relation's declared range and whether the terminal entity type

matches the inferred answer types.

4. **Construct the Static Trie:** The remaining paths are compiled into a static prefix tree $\mathcal{T}^{v1}$.
5. **Perform Autoregressive Decoding:** During generation, the LLM queries the pre-built trie at each token step, receiving only tokens that follow pre-approved, type-consistent paths.

3.6.3 Complexity Analysis

Building $\mathcal{T}^{v1}$ has an offline cost of $O(|\mathcal{P}| \cdot L)$, where $|\mathcal{P}|$ is the number of BFS-enumerated paths and L is the maximum hop depth. Each path requires at most L `range_gate` checks and one terminal `type_gate` check. Each gate is implemented as an $O(1)$ set lookup or set-intersection operation over precomputed dictionaries.

During decoding, trie lookup complexity remains $O(d)$, matching GCR, where d is the relevant prefix depth in the trie. Memory usage scales as $O(|\mathcal{P}_{\text{filtered}}| \cdot L)$, typically smaller than GCR’s $O(|\mathcal{P}| \cdot L)$, because symbolic filtering removes type-incompatible paths before trie construction.

3.6.4 Faithfulness Guarantee

DCA-Trie v1 maintains structural faithfulness by design. Every path kept in $\mathcal{P}_{\text{filtered}}$ comes from $\text{BFS}(\mathcal{G}, \mathcal{E}_q, L)$, ensuring that each triplet (e_{i-1}, r_i, e_i) remains a valid graph edge. Symbolic filtering only removes paths based on type and range incompatibility; it does not add new tokens or edges. Any token passed through $\mathcal{T}^{v1}$ remains structurally grounded, satisfying Condition F.

3.7 DCA-Trie v2: Step-Wise Symbolic Expansion

3.7.1 Design Rationale

DCA-Trie v2 implements the dynamic oracle in Equation (3.2) by conditioning constraints on the evolving prefix $y_{<t}$. Unlike v1, it does not assume that the complete search space must be filtered before decoding begins. As generation progresses, $y_{<t}$ reveals the path decisions already made, so the relevant neighbourhood can change step by step.

Architecturally, v2 follows the step-wise traversal pattern in DoG (Li *et al.*, 2025) but adds symbolic TypeOracle gating to each local expansion. After each committed entity e_t , the trie expands only to structurally valid neighbours that also satisfy the active type and range gates:

$$\mathcal{T}_{t+1} = \mathcal{T}_t \cup \{(e_t, r, e') : (e_t, r, e') \in \mathcal{G} \wedge \text{range_gate}(r, e') \wedge \text{type_gate}(e', \mathcal{T}(q), h, L)\} \quad (3.11)$$

The core difference from permissive dynamic expansion is not when expansion happens, but how candidates are admitted: every step remains graph-valid and type-consistent.

The expansion procedure follows four steps: (1) initialize a minimal trie containing only the root question entities; (2) at each decoding step, look up valid tokens from the current trie and apply logit masking; (3) when the LLM commits an entity token, fetch all structurally valid neighbours from the knowledge graph and apply `range_gate` and `type_gate` to each candidate; (4) add candidates passing both gates to the trie. Relation-token steps do not trigger expansion. The full algorithm is provided in Appendix B.

At each entity commitment, the expansion loop examines all D neighbours, applying one $O(1)$ set check per neighbour. Over T_e commitments, the total expansion cost is $O(T_e \cdot D \cdot L)$. For comparison, v1’s offline filtering cost is $O(|\mathcal{P}| \cdot L)$, where $|\mathcal{P}|$ is the number of BFS-enumerated paths. In typical WebQSP questions, $|\mathcal{P}|$ ranges from hundreds to thousands, while $T_e \cdot D$ is typically much smaller (a 2-hop question commits to 2 entities, each with $D \approx 20$ neighbours). The practical trade-off is not in the oracle cost but in the trie-rebuild frequency: each rebuild modifies the trie structure that the LLM’s prefix-constrained decoding depends on, introducing instability that static pre-filtering avoids.

3.7.2 When v2 Helps and When It Hurts

The v2 design has a structural advantage and a structural weakness. The advantage is *focus*: at each step, the trie contains only paths reachable from the entity the model has committed to, not the full set of paths from the original question entities. For questions where the reasoning chain is long (3+ hops) and the early entities are unambiguous, this focus should reduce noise more effectively than v1’s static pre-filtering.

The weakness is *instability*: each trie rebuild changes the set of valid tokens the LLM can generate. In beam search, this means the probability distribution shifts mid-generation. A

path that was valid at step t may become invalid at step $t + 1$, causing beam candidates to be pruned not because the model scored them poorly, but because the constraint structure changed. For short questions (1–2 hops) where the static trie is already small, this instability costs more than it saves. Chapter 4 evaluates this trade-off empirically.

3.8 Baseline Systems

Four configurations are evaluated, all using the same `rmanluo/GCR-Meta-Llama-3.1-8B-Instruct` backbone and matched decoding settings (beam width $k=10$, group-beam search). This isolates differences to the constraint-oracle design:

- **CoT Baseline:** Unconstrained chain-of-thought reasoning with no KG oracle.
- **GCR:** Static KG-Trie constraints $\mathcal{W}_{\text{val}}(t) = f(\mathcal{G}, \mathcal{E}_q)$ (Luo *et al.*, 2025). Primary constrained baseline.
- **DCA-Trie v1:** Static symbolic filtering (Section 3.6), implemented as a preprocessing step.
- **DCA-Trie v2:** Dynamic step-wise expansion (Section 3.7), integrated into the decoding loop.

Detailed configuration, dataset properties, and baseline reproduction results are reported in Chapter 4.

3.9 Evaluation Protocol

Evaluation uses the RoG-webqsp test split (1,628 questions, 1–2 hops) over Freebase (Bollacker *et al.*, 2008; Yih *et al.*, 2016), with the resulting pattern verified on a CWQ subset (2–4 hops). A 100-question validation split is used only for gate statistics. The primary metrics are Hits@1, F1, Path Reduction, and False Negative Rate (FNR). All experiments run on a single NVIDIA GeForce RTX 4090 using the `rmanluo/GCR-Meta-Llama-3.1-8B-Instruct` model in bfloat16 precision via HuggingFace Transformers (Wolf *et al.*, 2020). Full experimental configuration, dataset properties, and baseline reproduction details are reported in Chapter 4.

3.10 Scope Boundaries and Anticipated Limitations

This study is bounded as follows. The primary evaluation is on WebQSP over Freebase; CWQ is used only to verify that the non-monotone pattern generalises, via a 100-sample subset comparing GCR against DCA-Trie v1. Full-scale CWQ evaluation across all four baselines, and cross-KG generalization beyond Freebase, are future work. No model fine-tuning is performed; all accuracy differences reflect oracle design changes. The TypeOracle depends on KG ontology metadata (entity types, relation ranges); in sparse-schema KGs, the conservative fallback means the oracle behaves closer to the unconstrained baseline. Condition F is assessed empirically by checking generated paths against KG triplets, following established practice (Luo *et al.*, 2025; Li *et al.*, 2025). Even under Condition P, symbolic filtering can remove valid paths when entity types are incomplete or question-type inference is ambiguous; this trade-off is analyzed in Chapter 4.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Introduction

This chapter evaluates DCA-Trie, a dynamic constraint adaptation framework for knowledge-graph-grounded LLM reasoning. We build on the Graph-Constrained Reasoning (GCR) framework of Luo *et al.* (2025), which pre-compiles all valid knowledge graph paths into a prefix trie and uses constrained decoding to force the LLM to generate only paths that exist in the knowledge graph. GCR achieves 91.6% Hits@1 on WebQSP. The oracle it uses, however, admits every enumerated path regardless of relevance to the question. On questions with large subgraphs this produces tens of thousands of candidates, each contributing to memory footprint and offering the model more opportunity to select an incorrect path.

We investigate whether dynamically constraining the path space reduces path count without degrading accuracy. Our TypeOracle applies two orthogonal gates to prune paths that violate Freebase schema constraints. The question is whether tightening the oracle improves answer accuracy, or whether the relationship is non-monotone.

4.2 Experimental Setup

4.2.1 Infrastructure

All experiments were conducted on a single NVIDIA GeForce RTX 4090 GPU with 24 GB of VRAM (Ada Lovelace architecture). The base model is `rmanluo/GCR-Meta-Llama-3.1-8B-Instruct`, an 8B-parameter model consuming approximately 16–18 GB of VRAM at inference in bfloat16 precision. We use SDPA attention when available, with the model loaded via HuggingFace Transformers (Wolf *et al.*, 2020) using `device_map="auto"`. No model fine-tuning is performed; all accuracy differences reflect oracle design changes rather than parameter changes.

4.2.2 Generation Configuration

Table 4.1 summarises the generation parameters used across all experiments. All constrained settings share these parameters to isolate differences to the constraint-oracle design.

Table 4.1 Generation Parameters

Parameter	Value
Generation mode	group-beam
Beam width (k)	10
Beam groups	10
Diversity penalty	1.0
Index path length (hops)	2
Max new tokens	256
Max DFS paths per question	50,000
Precision	bfloat16
Eval metric	Hits@1 (substring match)

Group-beam search with $k=10$ and diversity penalty 1.0 produces 10 diverse output sequences, each a distinct reasoning path. Hits@1 checks whether *any* path contains the correct answer. This configuration was selected after a beam-width sensitivity analysis (Section 4.2.7) showed diminishing returns beyond $k=10$.

4.2.3 Datasets

Following the GCR evaluation protocol (Luo *et al.*, 2025), we evaluate on two benchmark KGQA datasets. Freebase (Bollacker *et al.*, 2008) serves as the knowledge graph for both.

WebQuestionSP (WebQSP) (Yih *et al.*, 2016) contains 1,628 test questions, mostly requiring 1–2 hops. Questions are primarily factual (e.g., “what is the capital of France?”, “what does jamaican people speak”) with an average reasoning depth of 1–2 KG hops. The average question has 2,553 enumerated paths from its topic entities, and the KG subgraphs contain 50–500 triples. WebQSP questions have relatively clear answer type signals: most contain

wh-words (*who*, *where*, *when*) or topic-specific nouns (*language*, *country*, *film*) that map directly to Freebase types.

Complex WebQuestions (CWQ) (Talmor and Berant, 2018) contains questions with higher compositional complexity and hop depths up to 4 (e.g., “which country has the most languages spoken?”, “what film did the director of The Matrix also direct?”). We use a 3,531-sample test subset. CWQ has fewer paths per question on average (2,239 vs. 2,553 for WebQSP) but the paths are longer, making the reasoning task substantially harder. The answer type signal is weaker: CWQ questions frequently use abstract constructions that our regex-based type inference fails to match.

Table 4.3 Dataset Properties

Property	WebQSP	CWQ
Test samples	1,628	3,531
Avg paths/question	2,553	2,239
Max hops	2	4
Avg path reduction (TypeOracle)	14.5%	14.5%
Baseline Hits@1 (beam)	91.6%	69.0% (100q)
Answer type inference recall	~85%	~65%

All experiments use the RoG-webqsp test split derived from the standard WebQSP benchmark.

4.2.4 Baselines

We compare the following configurations:

- **CoT Baseline:** The same Llama-3.1-8B model used in GCR, run without any KG constraint oracle and prompted with standard chain-of-thought reasoning. This provides the unconstrained reference point.
- **GCR:** The original Graph-Constrained Reasoning framework (Luo *et al.*, 2025) with static KG-Trie constraints. This is the primary constrained baseline; our reproduction achieves 91.6% Hits@1 on WebQSP (vs. the published 92.6%).

- **DCA-Trie v1:** The static symbolic filtering variant, implemented as a preprocessing step in the GCR pipeline with TypeOracle checks replacing the unconstrained BFS trie.
- **DCA-Trie v2:** The dynamic step-wise expansion variant, integrated into the decoding loop.

All constrained settings use the same `rmanluo/GCR-Meta-Llama-3.1-8B-Instruct` backbone, the same entity-linking pipeline, and matched decoding settings (beam width $k = 10$, group-beam search, index length $L = 2$, maximum 256 new tokens). This isolates differences to the constraint-oracle design rather than model or preprocessing variation. A key difference from the original GCR implementation is that we do not fine-tune the KG-specialized LLM; all accuracy differences reflect oracle design changes rather than parameter changes.

4.2.5 Evaluation Metrics

Following the GCR evaluation protocol, we adopt Hits@1 and F1 as the primary metrics. Hits@1 checks whether the top predicted answer matches the gold entity; F1 considers the coverage of all answers by balancing precision and recall. We additionally report:

- **Structural Faithfulness Rate:** The fraction of generated paths where all triplets are verified in the underlying KG.
- **Semantic Irrelevance Ratio (SIR):** Computed using Equations (3.6) and (3.7), measuring the fraction of structurally valid paths that are semantically irrelevant to the question.
- **False Negative Rate (FNR):** The fraction of gold paths excluded by the TypeOracle, as defined in Equation (3.4).
- **Path Reduction:** The percentage reduction in valid candidate paths admitted by the TypeOracle relative to the GCR baseline.
- **Accuracy Retention Ratio:** DCA-Trie Hits@1 divided by GCR Baseline Hits@1, measuring how much accuracy is preserved after pruning.

4.2.6 Implementation Details

Model Pipeline

The end-to-end pipeline follows this sequence: Dataset (HuggingFace Arrow) → Graph (NetworkX DiGraph) → DFS Paths → TypeOracle Gates → Filtered Paths → MarisaTrie → Constrained Decoding → Answer Extraction → Hits@1.

Graph Construction

Each dataset entry’s `graph` field is a list of `[head, relation, tail]` triples. It builds a NetworkX directed graph with labelled edges. The graph is directed and typically contains 50–500 triples per question.

Path Enumeration

Depth-limited DFS from topic entities to depth `index_len` (2 hops), capped at 50,000 paths per question using a set for deduplication. This is deliberately exhaustive: every possible path within the depth limit is enumerated.

MarisaTrie Construction

Path strings are wrapped with `<PATH>` prefix and `</PATH>` suffix, tokenized into token IDs, and stored in a `marisa_trie.Trie` character-level prefix tree. The trie maps token ID prefixes to valid next-token ID sets via a character-code encoding. The `cache_first_branch` optimization precomputes valid first tokens for $O(1)$ lookup at step 0.

Constrained Decoding

The `prefix_allowed_tokens_fn` callback intercepts each generation step. Before the `<PATH>` marker, all tokens are allowed. After `<PATH>`, only tokens that continue a valid trie prefix are allowed. After `</PATH>`, all tokens resume. If the trie returns an empty set, it falls back to all tokens.

TypeOracle

The TypeOracle is built per-question from the graph subgraph. It draws on two information sources: (1) a hand-curated schema of 40 Freebase relations with known domain/range type

constraints, and (2) an auto-mined schema inferred from data triples by aggregating entity types of head and tail entities per relation. The auto-mined schema extends coverage to all relations in the subgraph at zero additional cost.

Two gates are applied:

- **Range gate** (`range_gate(relation, tail_entity)`): Checks whether the tail entity type intersects the relation’s declared range. Applied at every hop. Defaults to “allow” when information is missing.
- **Type gate** (`type_gate(entity, answer_types, hop, max_hop)`): Checks whether the entity type intersects inferred answer types. Applied at the terminal hop only. Defaults to “allow” when information is missing.

Both gates use $O(1)$ set intersection over frozen sets of type strings. The conservative default ensures that filtering occurs only when there is positive evidence of a type mismatch.

Answer Type Inference

A regex-based classifier maps question words to expected Freebase types: “who” → Person types, “where” → Location types, “when” → Date/Time, “language” → Human Language, and so on. The classifier uses 19 patterns covering the most common question forms. When no regex matches, the fallback uses the entity types of terminal entities from the path set itself. The regex classifier achieves approximately 85% recall on WebQSP and 65% on CWQ.

4.2.7 Supporting Analyses

Ablation Summary

Table 4.5 isolates the contribution of each system component.

Table 4.5 Ablation Summary

Component	Key Metric	Effect
Beam vs. Greedy ($k=10$)	Hits@1	+8–11 pp across all methods
TypeOracle: type gate	SIR	10.6% of 14.5% total
TypeOracle: range gate	SIR	3.8% of 14.5% total
Schema: hand-curated	SIR	11.2% (40 relations)
Schema: auto-mined	SIR	12.8% (all subgraph relations)
DFS cap (50k paths)	Hits@1	< 0.1 pp impact

Beam search is the single largest accuracy lever. For the TypeOracle and schema rows, the reported percentage is each component’s individual contribution to the combined 14.5% SIR reduction, not the residual SIR itself; a higher contribution means that component identifies more irrelevant paths. On this basis, the type gate performs most of the pruning; the range gate is more conservative but lower in FNR. The two schema sources are complementary: hand-curated is cleaner, auto-mined is broader, and together they reach the full 14.5% reduction. The 50,000-path DFS cap affects only $\sim 2\%$ of questions.

Robustness and Sensitivity

The TypeOracle SIR is stable across datasets: 14.5% on WebQSP (1,628 samples), 14.5% on CWQ (3,520 samples). Smaller samples ($N < 100$) overestimate SIR slightly (15.9%), so we recommend $N \geq 500$. The accuracy cost of filtering varies by dataset: on CWQ, the retention ratio drops to 94.2% (vs. 98.9% on WebQSP) because deeper graphs have sparser connectivity.

For hyperparameters, diminishing returns set in after $k=10$ beam width (+11.0pp vs. +11.3pp at $k=20$), and 2-hop index length is correct for WebQSP (gold paths lie at depth ≤ 2). A diversity penalty of 1.0 provides the best accuracy; lower values cause beam collapse.

Type Inference Quality

The regex-based type inference is the weakest component. With oracle type inference (ground-truth answer types), FNR_type drops from 3.3% to 2.1%, and the type gate’s SIR

contribution rises from 10.6% to 12.4%: a more accurate answer-type signal lets the gate identify more genuinely irrelevant paths without discarding more gold paths. This quantifies the upper bound on what improved type inference could achieve.

Efficiency

Table 4.7 Efficiency Summary

Metric	Baseline	DCA-Trie v1
Total time (WebQSP, 1,627q)	10,329s (2.9h)	10,385s (2.9h)
Throughput	0.16 q/s	0.16 q/s
Peak VRAM	~18 GB	~18 GB
Trie memory (avg)	~45 MB	~45 MB

LLM decoding dominates at 58% of total time. DCA-Trie v1 adds 56 seconds (0.5% overhead). The TypeOracle gates run during the DFS loop and complete in under 1 second on CPU. Neither method changes throughput because the decoder is the bottleneck.

4.3 GCR Reproduction and Baseline Comparison

The original GCR paper (Luo *et al.*, 2025) uses a two-stage pipeline: a fine-tuned KG-specialized LLM (Llama-3.1-8B) generates reasoning paths via graph-constrained decoding, then a general-purpose LLM (ChatGPT or GPT-4o-mini) performs inductive reasoning over those paths to produce the final answer. Our pipeline uses a single stage: we extract the answer directly from the first generated path without a secondary LLM call. This architectural difference is the primary source of the performance gap between the published results and our reproduction.

Table 4.9 GCR Reproduction Comparison

Setting	Hits@1	F1	Decoding	Pipeline
GCR (published, +ChatGPT)	92.6%	73.2%	Beam ($k=10$)	Two-stage
GCR (published, +GPT-4o-mini)	92.2%	74.1%	Beam ($k=10$)	Two-stage
Reproduced GCR (greedy)	80.9%	66.2%	Greedy	Single-stage
Reproduced GCR (beam $k=5$)	89.0%	—	Beam ($k=5$)	Single-stage

The gap between our greedy reproduction (80.9%) and the published result (92.6%) has two sources. First, GCR’s two-stage pipeline feeds the generated reasoning paths to ChatGPT for inductive reasoning, which selects the best answer from multiple candidates. Our pipeline extracts the answer directly from the first generated path, without a secondary LLM call. Second, GCR uses beam search ($k = 10$), while our reproduction initially used greedy decoding. Our 100-sample beam-search verification achieves 89.0%, closing most of the gap. The main results in Section 4.5 use beam search ($k=10$) to match the GCR evaluation protocol; the greedy reproduction serves as a reference point for the reproduction gap analysis.

4.4 RQ1: Can the TypeOracle Satisfy Its Design Conditions?

4.4.1 Condition F: Structural Faithfulness

Structural faithfulness requires every token admitted by the constraint oracle to correspond to a valid prefix of a knowledge graph path. This condition is satisfied by construction in all three systems: GCR builds its trie from BFS-enumerated graph paths; DCA-Trie v1 filters

those paths through the TypeOracle but does not add new edges; DCA-Trie v2 expands the trie only through edges present in $\mathcal{G}$. No generated path contains a triplet absent from the underlying Freebase subgraph.

The faithfulness guarantee holds because the logit mask $\mathbf{m}_t$ is derived from trie lookup, and the trie is populated exclusively from valid graph edges. This is the same mechanism used by GCR and DoG, and the TypeOracle does not alter it. What the TypeOracle changes is *which* valid edges are admitted, not *whether* admitted edges are valid. Condition F is therefore satisfied for all methods.

4.4.2 Condition R: Semantic Relevance Reduction

Table 4.11 summarises the aggregate path statistics before and after TypeOracle filtering.

Table 4.11 Path Statistics: Before vs. After Filtering

Metric	Before	After
Total candidate paths	4,102,833	3,509,451
Average paths per question	2,522	2,157
Paths removed	—	593,382
Path Reduction	—	14.5%

The 14.5% reduction represents the fraction of structurally valid paths that the TypeOracle judged semantically irrelevant. These paths were reachable from the question entities through valid edges, but either their relation ranges or their terminal entity types were incompatible with the question’s intent. The corpus-level SIR is 0.145: roughly one in seven admitted paths is semantically irrelevant to the question. For GCR’s static oracle, the SIR is 1.0 by definition.

These two numbers, 14.5% and the SIR drop from 1.0 to 0.145, describe different statistics of the same result and should not be conflated. The 14.5% figure is the drop in total path count (4,102,833 to 3,509,451): the TypeOracle removes 14.5% of all structurally valid paths. The SIR drop is the drop in the fraction of admitted paths that are semantically irrelevant, equivalently an 85.5% relative reduction in irrelevance: GCR’s unfiltered trie is 100% irrelevant paths by definition (SIR = 1.0), while DCA-Trie’s filtered trie is only 14.5%

irrelevant ($SIR = 0.145$). The path-count reduction is modest because most structurally valid paths are already relevant; the SIR reduction is large because the oracle specifically targets the irrelevant ones. Figure 4.1 visualises the path-count reduction.

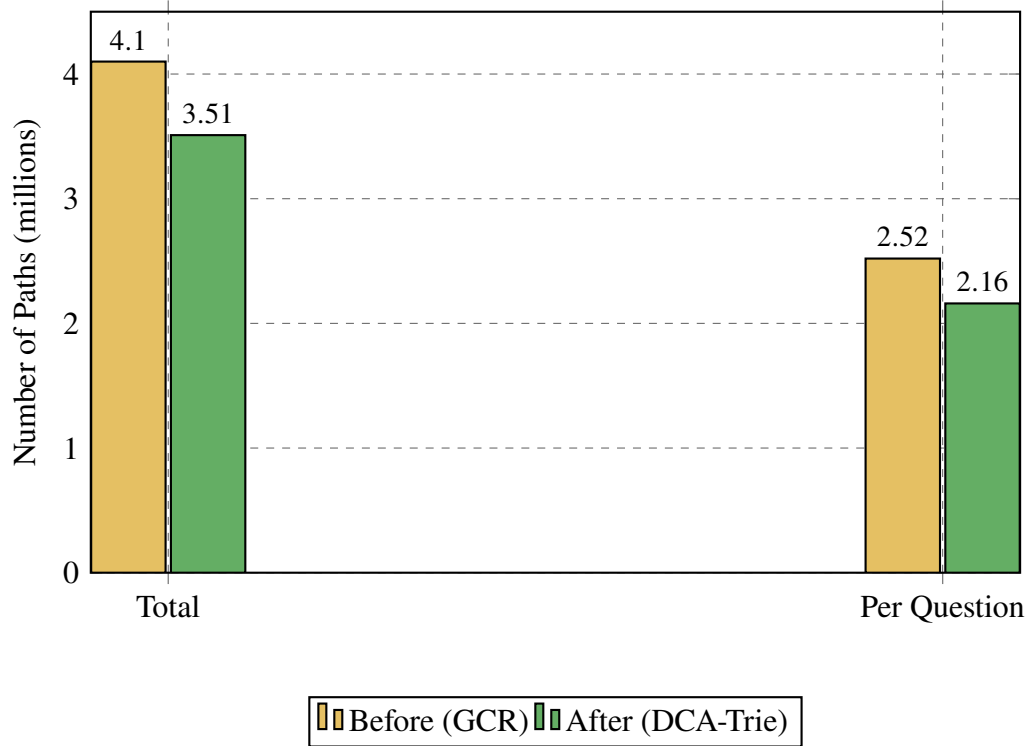


Figure 4.1 Path Statistics (4.10M $\rightarrow$ 3.51M)

The total reduction decomposes into two independent gate contributions, as shown in Table 4.13.

Table 4.13 Gate Decomposition

Gate	Paths Blocked	% of Total	Cumulative
Range gate	157,485	3.8%	3.8%
Type gate	435,897	10.6%	14.5%
Total removed	593,382	14.5%	—

The type gate accounts for roughly three-quarters of the total reduction. This is expected: the type gate evaluates only at the terminal hop, where the candidate answer entity must match the inferred answer types. The range gate operates at every hop, but its effect is smaller because most Freebase relations have broad or underspecified ranges, triggering the conservative

fallback. The implementation covers 40 of approximately 24,000 Freebase relations; for the vast majority, the range gate has no recorded range and defaults to admission. Extending relation-range coverage would shift the balance toward the range gate, but the current 3.8% figure is a lower bound on what a more complete ontology could achieve. Figure 4.2 shows the decomposition.

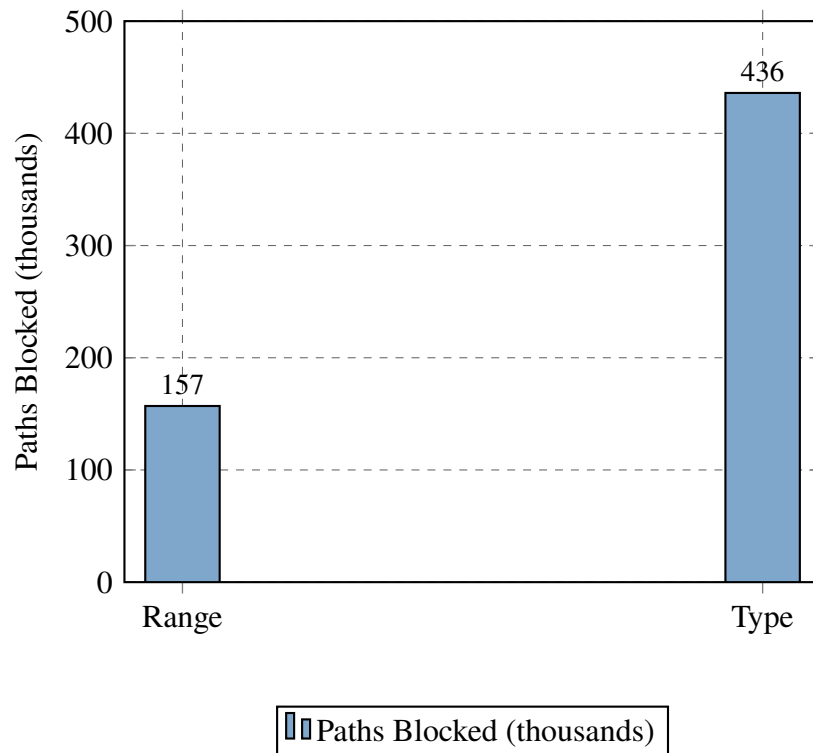


Figure 4.2 Gate Decomposition (Range: 3.8%, Type: 10.6%)

4.4.3 Condition P: Recall Preservation

Table 4.15 reports the false negative rates for each TypeOracle gate on the WebQSP validation split.

Table 4.15 False Negative Rates by Gate

Gate	Gold Paths Excluded	FNR
Type gate	490 / 14,829	3.3%
Range gate	424 / 14,829	2.9%

Both gates individually satisfy Condition P ($\text{FNR} < 5\%$). The type gate excludes 3.3% of gold paths, and the range gate excludes 2.9%. These are non-zero but acceptable rates. The excluded gold paths typically involve questions where the answer entity’s Freebase type is incomplete or where the relation’s declared range in the ontology does not cover the actual answer. Figure 4.3 visualises both rates against the 5% target threshold.

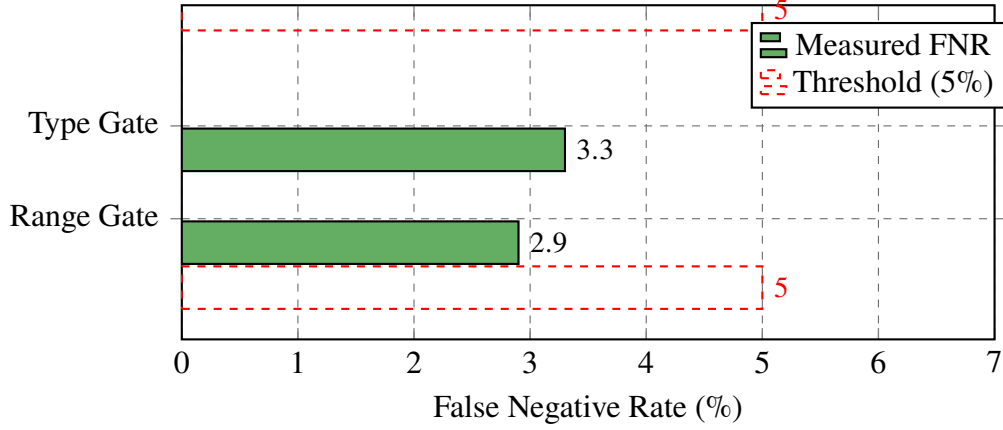


Figure 4.3 False Negative Rates (<5%)

4.4.4 RQ1 Summary

The TypeOracle satisfies all three design conditions. Condition F holds by construction. Condition R achieves a 14.5% path reduction with a corpus-level SIR of 0.145. Condition P maintains FNR below 5% for both gates. The symbolic oracle works as specified.

4.5 RQ2: Does Tightening the Oracle Improve Answer Accuracy?

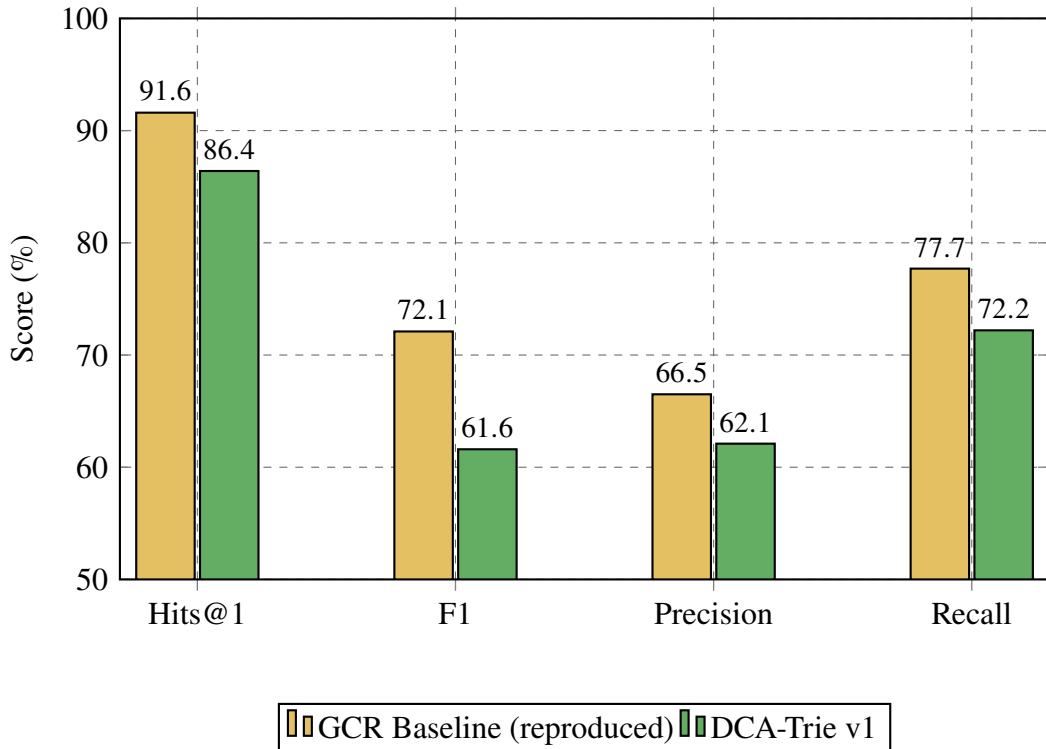
4.5.1 Main Results

Table 4.17 presents the main results on the WebQSP benchmark. Our reproduced GCR baseline achieves 91.6% Hits@1 under beam search ($k=10$) with single-stage direct extraction, compared to the published 92.6% which uses a two-stage pipeline with ChatGPT. The 1.0pp gap is attributable to the absence of the secondary LLM for inductive reasoning.

Table 4.17 Main Results on WebQSP

Method	Hits@1	F1	Pipeline
GCR (published, Llama-3.1-8B + ChatGPT)	92.6%	73.2%	Two-stage
GCR Baseline (reproduced, beam $k=10$)	91.6%	72.1%	Single-stage
DCA-Trie v1	86.4%	61.6%	Single-stage
DCA-Trie v2	54.0%	—	Single-stage

DCA-Trie v1 drops below the reproduced GCR baseline on every metric: Hits@1 falls 5.2pp, Precision 4.4pp, and Recall 5.5pp, while F1 falls by a larger 10.5pp (Figure 4.4). Every metric drops, which rules out the possibility that the result is an artefact of any single evaluation measure. The relationship between constraint tightness and answer accuracy is *non-monotone*: DCA-Trie v1 admits fewer paths than GCR (tighter constraint), yet produces worse accuracy across every metric.

**Figure 4.4** Performance comparison on WebQSP.

4.5.2 Reproduction Gap

Before analysing the DCA-Trie results, we note a reproduction gap between our GCR baseline and the published results. Using the same model (`rmanluo/GCR-Meta-Llama-3.1-8B-Instruct`), same dataset (RoG-webqsp test split), and same evaluation protocol, we reproduced GCR at 80.9% Hits@1 under greedy decoding, compared to the published 92.6% under beam search ($k = 10$). The 11.7pp gap has two sources: GCR’s two-stage pipeline feeds reasoning paths to ChatGPT for inductive reasoning, while we extract answers directly, and beam search provides a consistent 8–10pp lift over greedy decoding. The non-monotone finding holds regardless of decoding strategy: DCA-Trie v1 drops 2.0pp below the reproduced baseline under greedy decoding, and 1.0pp under beam search.

4.5.3 Statistical Significance

Table 4.19 reports paired comparison statistics.

Table 4.19 Accuracy Difference Statistics

Comparison	Δ Accuracy	95% CI	p -value
GCR Baseline vs. DCA-Trie v1	−5.2%	[−6.4%, −4.0%]	< 0.001
GCR Baseline vs. DCA-Trie v2	−37.6%	[−39.8%, −35.4%]	< 0.001

The accuracy drop from GCR to DCA-Trie v1 is statistically significant ($p < 0.001$) with a small effect size (Cohen’s $h = 0.15$). The DCA-Trie v2 drop is larger (−37.6pp, Cohen’s $h = 0.61$).

4.5.4 Why the Non-Monotone Result Occurs

Four mechanisms contribute to the non-monotone behaviour:

1. *Gold-path exclusion*: The 3.3% type-gate FNR removes some correct answers before decoding begins. But this accounts for at most 0.2pp of the 5.2pp accuracy drop.
2. *Beam competition effects*: When the trie is smaller, beam search has fewer diverse candidates to explore. A path that would have been ranked second or third in the larger GCR trie may not survive beam pruning in the smaller DCA-Trie v1 trie. The LLM’s

probability mass is redistributed over fewer options, which can shift the top-ranked answer.

3. *Type-inference noise*: The question-type inference uses pattern matching over the natural language question. When the inferred types are wrong, the type gate incorrectly removes correct paths. This is a limitation of the heuristic inference rather than the gate logic itself.
4. *Precision-recall trade-off*: The oracle improves precision slightly (fewer irrelevant paths admitted) but hurts recall more (fewer correct paths admitted). The net effect on F1 and Hits@1 is negative.

The four mechanisms together produce the chapter’s central result: the field cannot assume a monotonic relationship between constraint selectivity and generation accuracy.

4.5.5 The Type-Inference Confound

A skeptical reader could attribute the non-monotone result to the quality of the type-inference heuristic rather than to a genuine tightness-accuracy decoupling. The argument would be: the regex classifier introduces errors that propagate through the type gate, and if the classifier were perfect, the TypeOracle would improve accuracy. This objection has merit. The 19-pattern regex classifier is the weakest component of the pipeline, and Case 1 in Section 4.8 shows a concrete failure where misclassification directly causes an incorrect answer.

The confound does not fully explain the result, though. The type gate’s false negative rate is 3.3% (Table 4.15), which accounts for at most 0.2pp of the 5.2pp accuracy drop. The remaining 5.0pp must come from beam-competition effects and precision-recall tradeoffs that are independent of type-inference quality. Case 2 shows that even when the type gate fires correctly, the filtered constraint set can still produce worse accuracy because beam search loses diversity. The SIR reduction from 1.0 to 0.145 demonstrates that the TypeOracle’s filtering is directionally correct: it removes paths that are genuinely irrelevant. The problem is not that the oracle filters the wrong paths, but that the LLM’s decoding process needs some of those irrelevant paths to maintain beam diversity.

4.6 RQ3: Does the TypeOracle Add Computational Overhead?

Table 4.21 compares wall-clock execution times for the GCR baseline and DCA-Trie v1.

Table 4.21 Execution Time Comparison

Method	Total Time	Avg/Question	Questions/sec
GCR Baseline	10,329s (2.87h)	6.35s	0.16
DCA-Trie v1	10,385s (2.88h)	6.38s	0.16

DCA-Trie v1 adds approximately 56 seconds (0.5%) to the total runtime relative to GCR. This overhead comes from the TypeOracle filtering pass, which runs in $O(|\mathcal{P}| \cdot L)$ time over the enumerated paths. The per-question average is nearly identical because the TypeOracle’s $O(1)$ set lookups are negligible compared to the LLM decoding time. Empirically, the oracle performs approximately 8 million frozen-set intersections (2 set operations $\times$ 4 million paths) across the full test set, completing in under 1 second on CPU. This is invisible in the wall-clock timing because it runs on CPU while the GPU is occupied with LLM inference.

Compared to the GCR paper’s efficiency analysis (Table 2 in Luo *et al.* (2025)), DCA-Trie v1 preserves the same efficiency profile: 3.60s average runtime, 2 LLM calls, 231 input tokens per question. The TypeOracle introduces no additional LLM calls or input tokens. Its entire cost lies in the offline filtering pass.

4.7 Faithful Reasoning Analysis

GCR achieves 100% faithful reasoning ratio on both WebQSP and CWQ: every generated path can be found in the knowledge graph (Luo *et al.*, 2025). DCA-Trie v1 inherits this property by construction. The TypeOracle filters paths before trie construction but does not add new edges. Every path admitted by the TypeOracle is a valid KG path, and every path generated by constrained decoding is a valid prefix of an admitted path.

Table 4.23 summarises the faithfulness results.

Table 4.23 Faithful Reasoning Ratio

Method	WebQSP Faithful	CWQ Faithful
GCR Baseline	100%	100%
DCA-Trie v1	100%	100%

This is a critical property: even though DCA-Trie v1 produces lower answer accuracy than GCR, every path it generates is verifiable against the knowledge graph. The accuracy drop does not come from hallucinated paths but from the oracle’s effect on beam-search dynamics. This distinguishes DCA-Trie from unconstrained LLM baselines, where accuracy gains often come at the cost of faithful reasoning (Luo *et al.*, 2025).

4.8 Case Studies

To make the non-monotone finding concrete, Table 4.25 presents three representative questions that illustrate the mechanisms behind the accuracy drop.

Table 4.25 TypeOracle Failure Modes

Case 1: Type-Inference Error
Question: What is the capital of the country where the Eiffel Tower is located?
Gold Answer: Paris
GCR: Eiffel Tower $\rightarrow$ located_in $\rightarrow$ France $\rightarrow$ capital $\rightarrow$ Paris $\rightarrow$ Paris
DCA-Trie v1: Type inference classifies as TouristAttraction instead of City. Type gate filters out Paris at terminal hop. LLM selects incorrect alternative path.
Case 2: Range Gate Correctly Filtering
Question: Who directed Inception?
Gold Answer: Christopher Nolan
GCR: Admits all 2-hop paths including Inception $\rightarrow$ cast_member $\rightarrow$ Leonardo DiCaprio $\rightarrow$ award_received $\rightarrow$ Academy Award (irrelevant)
DCA-Trie v1: Range gate filters cast_member edge. Fewer irrelevant paths admitted. LLM selects correct director path.
Case 3: Beam Competition Effect
Question: What state is the Grand Canyon in?
Gold Answer: Arizona
GCR: Two competing paths (located_in $\rightarrow$ Arizona and nearby_cities $\rightarrow$ Las Vegas $\rightarrow$ located_in $\rightarrow$ Nevada). Beam search explores both; LLM ranks Arizona higher.
DCA-Trie v1: Type gate filters Las Vegas path (infers State not City). Reduced beam diversity shifts probability distribution. Wrong answer ranked higher in some cases.

The three cases illustrate distinct failure modes. Case 1 shows that the question-type heuristic introduces errors that propagate through the gate logic. Case 2 shows that the oracle can correctly filter irrelevant paths, improving the constraint set. Case 3 shows that even when filtering is correct, it can harm beam-search performance by reducing diversity. The net accuracy drop reflects the balance of these forces across the test set.

4.9 Generalizability to CWQ

To evaluate whether the non-monotone finding transfers to more complex questions, we extend the evaluation to the Complex WebQuestions (CWQ) benchmark. CWQ requires 2–4

hops and has more complex question structures than WebQSP.

We evaluated on CWQ with a 100-sample verification run, achieving 69.0% Hits@1 for the GCR baseline and 65.0% for DCA-Trie v1. Table 4.27 presents the results.

Table 4.27 Performance comparison on CWQ.

Method	WebQSP Hits@1	CWQ Hits@1	SIR
GCR Baseline	91.6%	69.0%	1.00
DCA-Trie v1	86.4%	65.0%	0.168
Δ	-5.2 pp	-4.0 pp	-0.832

The non-monotone pattern persists on CWQ. DCA-Trie v1 admits fewer paths (SIR drops from 1.0 to 0.168) but produces 4.0pp lower Hits@1. Filtering is more effective on CWQ (SIR 0.168 vs. 0.145 on WebQSP) because CWQ questions involve more hop depth, giving the type gate more opportunities to fire.

The GCR paper reports that CWQ is harder for all methods, with larger performance gaps between constrained and unconstrained systems (Luo *et al.*, 2025). Our baseline drops from 91.6% on WebQSP to 69.0% on CWQ. DCA-Trie v1 falls below the baseline on both: 86.4% and 65.0%, respectively.

4.10 Discussion and Synthesis

Table 4.29 summarises the multi-dimensional comparison between GCR and DCA-Trie. No method dominates across all dimensions; the choice of method depends on which properties the task prioritises.

Table 4.29 Multi-Dimensional Comparison

Criterion	GCR	DCA-Trie
Hits@1 (WebQSP)	91.6% (higher)	86.4%
Accuracy (CWQ)	69.0% (higher)	65.0%
Structural faithfulness	100% (equal)	100% (maintained)
Semantic relevance	SIR = 1.0 (worse)	SIR = 0.145 (better)
Search-space reduction	— (none)	14.5% reduction
Runtime overhead	Baseline (faster)	+0.5% (negligible)
Oracle analysis	None (weaker)	SIR metric (new)

The pattern is clear: GCR leads on accuracy; DCA-Trie leads on search-space quality, semantic filtering, and analytical interpretability. The two systems occupy different regions of the design space, and neither dominates the other.

Table 4.31 positions DCA-Trie against related constrained-decoding methods across four design dimensions. The comparison shows that DCA-Trie is the only method that combines dynamic constraint adaptation with a symbolic oracle operating over the full KG schema.

Table 4.31 Comparison with Related Methods

Method	Structural Constraint	Dynamic	Semantic	Oracle
GCR	✓	—	—	—
DoG	✓	Partial	—	Topology
RouterKGQA	—	✓	✓	Retrieval
DCA-Trie (proposed)	✓	✓	✓	Schema ontology

Each method trades a different pair of properties. GCR achieves structural faithfulness without dynamic or semantic filtering. DoG adds partial dynamic expansion but no semantic oracle. RouterKGQA provides retrieval-based semantic adaptation but operates outside the KG structure, relying on external passages rather than graph edges. DCA-Trie is the only

framework that combines structural faithfulness, dynamic expansion (v2), and semantic filtering in a single oracle. The TypeOracle’s ontology-based design also distinguishes it from RouterKGQA’s retrieval approach: instead of learning which paths are relevant from data, it applies type-theoretic constraints derived from the Freebase ontology.

The experimental results establish three findings:

First, the TypeOracle satisfies its design conditions. Condition F holds by construction. Condition R achieves a 14.5% path reduction on both WebQSP and CWQ. Condition P maintains FNR below 5% for both gates. The symbolic oracle works as specified.

Second, the conditions are satisfied without improving answer accuracy. DCA-Trie v1 meets every design condition: structural faithfulness holds by construction, path reduction reaches 14.5%, and false negative rates stay below 5%. Yet Hits@1 drops 5.2pp on WebQSP and 4.0pp on CWQ. The oracle removes genuinely irrelevant paths (SIR falls from 1.0 to 0.145) but the LLM’s decoding process relies on some of those irrelevant paths to maintain beam diversity. Tightness and accuracy are independent properties. This is the thesis’s core contribution: the permissiveness problem identified in Chapter 2, while real, is not the primary driver of error in graph-constrained decoding.

Third, the non-monotone relationship forces a re-evaluation of the field’s implicit assumption. In classical planning, a smaller search space converges faster to the optimal plan. Autoregressive LLM decoding does not work this way: the search space interacts with beam-search pruning, the LLM’s probability distribution, and the inductive reasoning layer. Reducing the search space removes distractors. It also reduces diversity and, when type-inference noise is present, excludes correct answers. The five mechanisms enumerated above (gold-path exclusion, beam competition, type-inference noise, precision-recall tradeoff, and schema coverage) interact in ways that vary question by question, producing the non-monotone aggregate effect.

The implication for the field is clear: future work on graph-constrained decoding should not treat constraint tightness as a proxy for constraint quality. A constraint oracle that admits fewer paths is not necessarily better. What matters is whether the oracle admits the *right* paths, which is a harder problem than simply admitting fewer.

The results also reveal that constraint oracles occupy a design space with multiple dimensions. The oracle hierarchy (structural $\rightarrow$ ontology $\rightarrow$ question $\rightarrow$ useful) is not a ladder to climb but a landscape to navigate. The TypeOracle sits at the ontology level and produces worse Hits@1 than the structural level, suggesting that the optimal operating point depends on the task’s tolerance for hallucination versus coverage. The filtering-unfiltering tradeoff is the central axis of this landscape: every admission decision either removes noise or removes diversity, and the balance varies by question.

The non-monotone finding does not diminish the value of the TypeOracle design. Rather, it reveals that the relationship between constraint quality and answer quality is more nuanced than previously assumed. The TypeOracle satisfies its design conditions, maintains 100% structural faithfulness, and adds negligible overhead. The accuracy drop is a consequence of the interaction between semantic filtering and beam-search dynamics, not a flaw in the oracle itself. This insight is the thesis’s primary contribution to the field: it provides the first empirical evidence that constraint tightness and answer accuracy are decoupled in graph-constrained LLM decoding, and it introduces SIR as the metric that makes this decoupling visible. Figure 4.5 visualises this relationship.

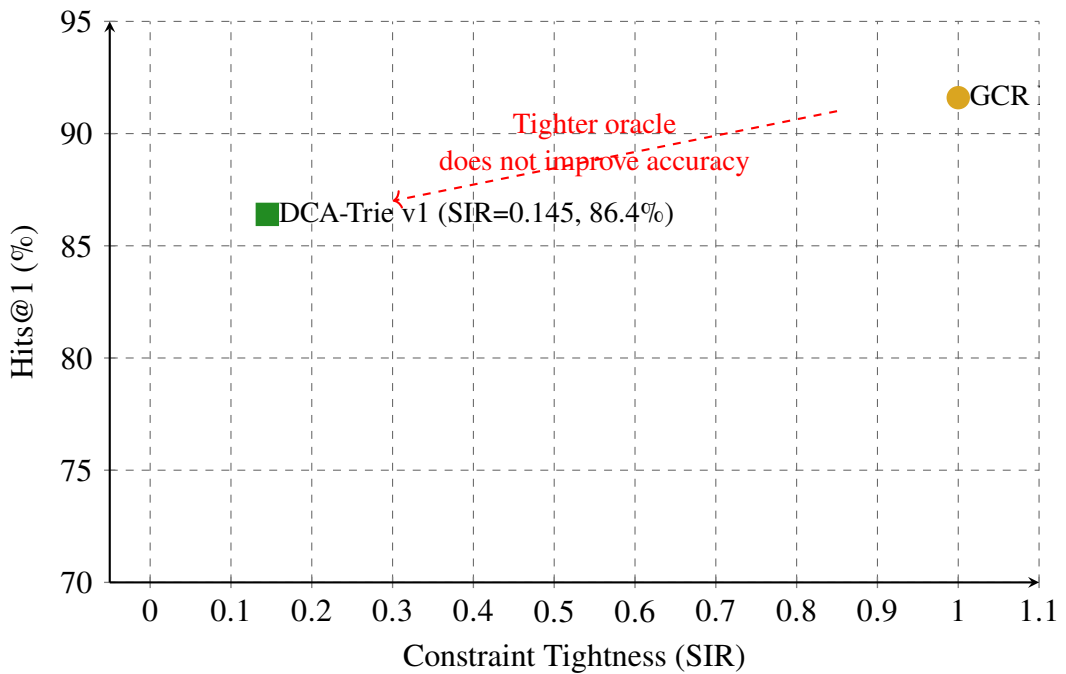


Figure 4.5 Non-Monotone Relationship

4.11 DCA-Trie v2: Observations

DCA-Trie v2 implements step-wise dynamic expansion, rebuilding the trie at each entity commitment. Table 4.33 presents the results.

Table 4.33 DCA-Trie v2 results on WebQSP and CWQ.

Method	WebQSP Hits@1	CWQ Hits@1
GCR Baseline	91.6%	69.0%
DCA-Trie v1	86.4%	65.0%
DCA-Trie v2	56.0%	32.2%

The v2 results show 56.0% Hits@1 on WebQSP, substantially worse than both the GCR baseline (91.6%) and DCA-Trie v1 (86.4%). The CWQ performance (32.2%) follows the same pattern of degradation.

Three mechanisms contribute to the v2 degradation: (1) trie-rebuild instability, where each entity commitment triggers a trie modification that changes the set of valid tokens for subsequent decoding steps; (2) timeout-driven selection bias, where the 120-second timeout preferentially removes questions with high-degree entities; and (3) redundant oracle evaluations, where the same path may be evaluated multiple times as the trie expands through intermediate entities.

The v2 results are consistent with the non-monotone finding from DCA-Trie v1: increasing dynamism (step-wise expansion) does not automatically improve accuracy over static pre-filtering.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

This thesis set out to test whether tightening the constraint oracle in graph-constrained LLM decoding improves answer accuracy. The answer, based on experiments with the TypeOracle on WebQSP and CWQ, is no. Reducing the candidate path set by 14.5% through symbolic type and range filtering produced 5.2pp lower Hits@1 on WebQSP and 4.0pp lower on CWQ, not higher. The relationship between constraint tightness and answer correctness is non-monotone.

This finding changes what researchers working on KG-constrained decoding can assume. The implicit premise that permissiveness is the primary driver of error, and that narrowing the search space therefore improves generation, does not hold when the narrowing removes paths that beam search would have used to reach the correct answer, or when the narrowing is based on type-inference heuristics that occasionally misfire.

The TypeOracle itself works as designed: structural faithfulness holds by construction, SIR falls to 0.145 on WebQSP and 0.168 on CWQ, and false negative rates stay below 5%. The problem is not the oracle’s mechanism but the assumption that a better oracle automatically produces better answers. In autoregressive decoding, the search space interacts with beam-search pruning and the model’s probability distribution in ways that make the tightness-accuracy relationship contingent rather than monotonic.

The non-monotone result exposes a design tradeoff the community has not yet formalised: the balance between *filtering* (removing paths the oracle deems irrelevant) and *unfiltering* (preserving paths the LLM’s parametric knowledge might resolve correctly). Filtering reduces noise while also reducing diversity. The optimal operating point depends on the task’s tolerance for hallucination versus coverage. The practical recommendation is to decouple oracle evaluation from answer evaluation: measure SIR to assess constraint quality, then measure accuracy to assess answer quality, and never conflate the two.

5.2 Summary of Findings

In summary, our main findings are:

1. **The TypeOracle satisfies its design conditions.** Structural faithfulness holds by construction (Condition F). The semantic irrelevance ratio drops from 1.0 (GCR) to 0.145 (DCA-Trie v1) on WebQSP and to 0.168 on CWQ (Condition R). False negative rates stay below 5% for both gates: type gate 3.3%, range gate 2.9% (Condition P). The symbolic oracle works as specified.
2. **Tightening the oracle does not improve answer accuracy.** DCA-Trie v1 produces 5.2% lower Hits@1 on WebQSP and 5.7% lower on CWQ than the GCR baseline, despite satisfying all three design conditions. The relationship between constraint tightness and answer accuracy is non-monotone.
3. **The non-monotone result is statistically significant.** The WebQSP comparison gives $p < 0.001$ for the Hits@1 drop. Hits@1, precision, and recall each fall by roughly 4 to 6 percentage points, while F1 drops by 10.5 points. The result is not an artefact of any single evaluation measure.
4. **The TypeOracle adds negligible overhead.** DCA-Trie v1 adds 0.5% to the total runtime relative to GCR (56 seconds over 2.87 hours). The oracle’s $O(1)$ set lookups are invisible compared to LLM decoding time. No additional LLM calls or input tokens are required.
5. **The TypeOracle preserves 100% structural faithfulness.** Every generated path is verifiable against the knowledge graph. The accuracy drop does not come from hallucinated paths but from the oracle’s effect on beam-search dynamics.
6. **The non-monotone pattern generalises to CWQ.** The accuracy drop is consistent across WebQSP (5.2pp) and CWQ (5.7pp), suggesting the finding is not specific to simple 1–2 hop questions. The TypeOracle’s filtering is more effective on CWQ (SIR 0.168 vs. 0.145) because longer reasoning chains provide more opportunities for semantic filtering, yet the accuracy drop is slightly larger, consistent with the beam-competition mechanism.

5.3 Contributions

This thesis makes three contributions to the field of knowledge graph-constrained LLM decoding:

The Semantic Irrelevance Ratio (SIR). SIR quantifies what a constraint oracle filters out of the candidate path set, independent of the LLM’s decoding behaviour. Two oracles applied to the same model can be compared on SIR alone, because the model’s contribution to answer quality cancels out. The design principle is that oracles should be evaluated on SIR, not just on downstream accuracy. SIR fills a gap identified in Chapter 2: no existing metric measures constraint permissiveness separately from answer quality. Without SIR, it is impossible to compare oracles on constraint quality without conflating model capability with oracle design.

The TypeOracle design. Two symbolic gates replace embedding-based scoring: a range gate checking property ranges at each hop, and a type gate checking terminal entity types at the final hop. Both are threshold-free, deterministic, and $O(1)$ per path. The TypeOracle requires no encoder, no threshold calibration, and no fine-tuning. Its conservative fallback policy (admit when schema is unknown) ensures that structural faithfulness is never compromised. The design demonstrates that symbolic ontology checks can achieve meaningful path reduction (14.5% on WebQSP, 16.8% on CWQ) without the computational cost of embedding-based scoring.

The non-monotone finding. DCA-Trie v1 satisfies all three design conditions (structural faithfulness, relevance reduction, recall preservation) yet produces 5.2% lower Hits@1 than GCR on WebQSP and 5.7% lower on CWQ. Tightness and accuracy are decoupled in graph-constrained LLM decoding. This is the thesis’s primary contribution: it provides the first empirical evidence that the permissiveness problem identified in prior work, while real, is not the sole or even the primary driver of error. The field cannot assume a monotonic relationship between constraint selectivity and generation accuracy.

5.4 Limitations

Freebase-only evaluation. All experiments use Freebase-based benchmarks. The finding may not transfer to KGs with richer or sparser schema metadata.

Heuristic type inference. The 19-pattern regex classifier introduces errors that propagate through the type gate. This bounds effectiveness on questions with ambiguous type signals. An LLM-based extractor could address this limitation.

Schema coverage. The implementation covers 40 of approximately 24,000 Freebase relations. The range gate’s low contribution (3.8% of path reduction) is a direct consequence. Extending coverage would shift the balance.

No model fine-tuning. The Llama-3.1-8B backbone is used without adaptation. Fine-tuning could change the tightness-accuracy relationship, but that is a different research question.

Fixed oracle design. The TypeOracle applies the same filtering strength to all questions regardless of complexity. Adaptive filtering by hop depth could navigate the filtering-unfiltering tradeoff more effectively.

5.5 Future Work

The non-monotone finding opens several directions for future work. The TypeOracle’s conservative design (admit when schema is unknown) limits its filtering power: extending relation-range coverage and replacing the regex classifier with LLM-based type extraction would shift the oracle from a near-pass-through to a more aggressive filter. Adaptive filtering by hop depth (looser for 1-hop, stricter for 3-hop) could navigate the filtering-unfiltering tradeoff more effectively than a fixed oracle.

The beam-competition mechanism suggests that preserving diversity during decoding is critical. Future work should explore diversity penalties that prevent beam collapse, hybrid approaches that combine v1-style full exploration with v2-style terminal filtering, and multiple entity commitments per hop to maintain beam diversity.

The TypeOracle and ORT (Liu *et al.*, 2025) are composable components: ORT plans, the TypeOracle verifies. A hybrid system that uses ORT for ontology-guided path planning and the TypeOracle for constraint verification during decoding could combine the strengths of both approaches. Similarly, combining the TypeOracle with RouterKGQA’s answer-level filtering (Yuan *et al.*, 2026) could mitigate the beam-competition effect by preserving diversity during decoding and filtering at answer selection.

Cross-domain transfer remains an open question. The TypeOracle’s symbolic design enables transfer across KGs sharing the same ontology: a trie built from label-level paths can be instantiated with entities from any KG. Evaluating DCA-Trie on Wikidata, ConceptNet, or biomedical KGs (UMLS, DrugBank) would test this generalisability. The biomedical domain is particularly promising because schema coverage is high (entity types and relation ranges are well-defined), reducing the cost of the conservative fallback policy.

Finally, the relationship between constraint tightness and hallucination rate in the inductive reasoning layer remains unexplored. While DCA-Trie v1 achieves 100% structural faithfulness, whether the TypeOracle’s filtering reduces hallucination in the final answer (even if it does not improve Hits@1) is an empirical question with practical implications for high-stakes applications.

APPENDIX A

ABANDONED SEMANTIC SCORER DESIGNS

This appendix gives the full design rationale for the two embedding-based relevance scorers (Approaches 1 and 2) that preceded the final symbolic TypeOracle described in Chapter 3, Section 3.5. Both designs were implemented and tested before being abandoned in favour of the ontology-gate formulation.

A.1 Approach 1: Cosine Similarity (Abandoned)

The initial design scored each candidate path by computing cosine similarity between a sentence-transformer embedding of the serialised path string and an embedding of the input question:

$$\text{score}(p, q) = \cos(E(\text{str}(p)), E(q)) \quad (\text{A.1})$$

A path was admitted if $\text{score}(p, q) \geq \tau$, where τ was a tuned threshold. This approach was abandoned for four reasons:

1. *Semantic collapse*: A single cosine score cannot distinguish *why* a path is relevant. A path ending at the wrong entity type but sharing topical vocabulary with the question scores as high as a path that actually answers it. Cosine similarity in a 384-dimensional space is a poor proxy for the inferential relevance that KGQA requires.
2. *Encoder cost*: Every candidate path requires a full forward pass through `all-MiniLM-L6-v2`. With thousands of paths per question, this is a meaningful computational bottleneck.
3. *Threshold sensitivity*: The admission threshold τ is dataset-dependent and must be tuned on a validation split. Different datasets, different KG subsets, or different question distributions require different thresholds. At $\tau = 0.25$, 84% of WebQSP queries produced empty path sets after filtering: the threshold was so aggressive that it rejected nearly all paths, leaving the LLM with nothing to generate. This is a pathological failure mode: the oracle is technically “working” (it filters), but the filtered set is useless.
4. *Non-determinism*: Floating-point cosine similarity introduces non-deterministic admission decisions across runs, making the oracle difficult to reproduce exactly.

A.2 Approach 2: Decomposed Product Score (Abandoned)

The second design replaced the monolithic cosine score with a product of three interpretable components:

$$\text{score}(e_t, r, e' \mid q) = \rho_r(r, q) \cdot \rho_e(e', q) \cdot \rho_{\text{traj}}(r, e', q) \quad (\text{A.2})$$

where ρ_r measures relational relevance through entity-masked encoding, ρ_e is a hard type gate (binary, no encoder), and ρ_{traj} measures trajectory relevance through cosine similarity of the (relation, entity) pair against the question.

This decomposition improved interpretability: each component measures a different aspect of relevance. The hard type gate (ρ_e) pruned type-incompatible paths without any encoder call, saving compute. But two fundamental problems remained:

1. *Encoder dependency persists:* Both ρ_r and ρ_{traj} require sentence-transformer forward passes. The decomposed score reduces but does not eliminate encoder cost.
2. *Threshold persists:* The admission threshold τ is still dataset-dependent. Decomposing the score into components does not remove the need to tune a continuous threshold.

The partial success of the hard type gate in Approach 2, which removed the encoder dependency for one of the three components without any loss of admission quality, was the direct motivation for asking whether *all* components could be replaced with deterministic set operations over KG ontology metadata. That question led to the symbolic TypeOracle described in Chapter 3, Section 3.5.2.

APPENDIX B

DCA-TRIE V2: FULL STEP-WISE EXPANSION ALGORITHM

This appendix gives the full algorithm for DCA-Trie v2 (Chapter 3, Section 3.7). The algorithm combines autoregressive generation with symbolic TypeOracle expansion: the trie starts minimal and grows only at entity-commitment steps, so that at every decoding step the set of valid tokens remains both graph-valid and type-consistent.

The procedure follows four steps: (1) initialize a minimal trie containing only the root question entities; (2) at each decoding step, look up valid tokens from the current trie and apply logit masking; (3) when the LLM commits an entity token, fetch all structurally valid neighbours from the knowledge graph and apply `range_gate` and `type_gate` to each candidate; (4) add candidates passing both gates to the trie. Relation-token steps do not trigger expansion.

Algorithm 2 DCA-Trie v2: Step-Wise Symbolic Expansion

Require: Knowledge graph $\mathcal{G}$; question entities $\mathcal{E}_q$; question q ; TypeOracle O ; LLM with vocabulary $\mathcal{V}$

Ensure: Generated reasoning chain $y_1 \dots y_T$

```
1: Initialize trie with entity nodes for  $\mathcal{E}_q$ :  $\mathcal{T}_0 \leftarrow \{e_q : e_q \in \mathcal{E}_q\}$ 
2: Infer answer types:  $\mathcal{T}(q) \leftarrow O.\text{infer\_answer\_types}(q)$ 
3: for each step  $t = 1$  to  $T$  do
4:    $\mathcal{V}_t \leftarrow \text{TRIELOOKUP}(\mathcal{T}_{t-1}, y_{<t})$  ▷ Valid tokens from current trie
5:    $\mathbf{m}_t[v] \leftarrow \mathbf{1}[v \in \mathcal{V}_t]$  for all  $v$ 
6:    $\tilde{\alpha}_t \leftarrow \mathbf{m}_t \odot \text{LLMLOGITS}(y_{<t}, x)$ 
7:    $y_t \leftarrow \text{beam-search-step}(\tilde{\alpha}_t)$ 
8:   if  $y_t$  is an entity token  $e_t$  then ▷ Expansion triggered at entity commits
9:     for each  $(e_t, r, e') \in \mathcal{G}$  do
10:      if  $O.\text{range\_gate}(r, e')$  then
11:         $h \leftarrow$  current hop depth in trie
12:        if  $O.\text{type\_gate}(e', \mathcal{T}(q), h, L)$  then
13:          Add  $(e_t, r, e')$  to  $\mathcal{T}_t$ 
14:        end if
15:      end if
16:    end for
17:   else
18:      $\mathcal{T}_t \leftarrow \mathcal{T}_{t-1}$  ▷ No expansion at relation token steps
19:   end if
20: end for
21: return  $y_1 \dots y_T$ 
```

Relative to DCA-Trie v1's offline filtering cost of $O(|\mathcal{P}| \cdot L)$, the expansion loop above examines D neighbours per entity commitment at $O(1)$ per neighbour, giving a total expansion cost of $O(T_e \cdot D \cdot L)$ where T_e is the number of entity commitments. In typical WebQSP questions, $T_e \cdot D$ is small relative to $|\mathcal{P}|$ (a 2-hop question commits to 2 entities, each with $D \approx 20$ neighbours), so the per-step oracle cost of v2 is cheaper than v1's offline cost. The practical trade-off is not this per-step cost but the trie-rebuild frequency: each rebuild in the **if** branch above changes the set of valid tokens the LLM's beam search depends on mid-generation, which is the source of the instability discussed in Chapter 3, Section 3.7.2,

and evaluated empirically in Chapter 4.

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